

# Operational Soldering from Relativistic Massive-State Geometry: An Inverse Reconstruction

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**Abstract** We formulate an inverse reconstruction of local Lorentzian geometry from scalar incidence data associated with relativistic massive generators. The companion flat-space analysis supplies the standard massive-generator geometry and connected Lorentz observer structure. Here an internal rank-four Lorentz bundle  $V \rightarrow M$  is initially independent of the tangent bundle  $TM$ , and no spacetime metric or solder form is assumed. Given a scalar generator–displacement response  $\Theta_x(\Pi, \xi)$  on the positive massive-generator cone and infinitesimal tangent displacements, additive composition, positive homogeneity, continuity, smooth base-point dependence, and operational faithfulness yield a unique bilinear extension. Finite-dimensional duality then reconstructs a unique solder map  $e_x : T_x M \rightarrow V_x$ ; faithfulness and equal rank make it an isomorphism, and the internal Lorentz form induces the spacetime metric  $g = e^* \eta$ . A calibrated fixed-mass shell is transferred to the standard cotangent mass shell, whereas a response known only up to positive local rescaling determines only the conformal metric class. As a separate downstream result, imposing an auxiliary torsion-free branch, metric ADM phase space, the standard Lorentzian hypersurface-deformation algebra, and a restricted separable quadratic scalar constraint yields the ADM/Einstein branch by off-shell closure. This gravitational completion is conditional; the principal result is the inverse operational reconstruction of Lorentzian soldering from the positive massive-generator sector.

**Keywords:** operational reconstruction; Lorentzian soldering; motion–staticity representation; massive generator geometry; hypersurface deformations; canonical gravity

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# 1 Introduction

The metric and coframe are normally introduced as basic geometrical fields of relativistic space-time. An inverse question is whether part of this geometry can instead be characterized by physically accessible relations that do not already presuppose the structures to be reconstructed. Several established approaches address related questions. The Ehlers–Pirani–Schild program reconstructs compatible conformal and projective structures from light propagation and freely falling trajectories [4]; the assumptions underlying this construction, including possible roles of torsion and non-metricity, have also been revisited [28]. Under suitable causality conditions, chronological or causal relations determine conformal spacetime structure up to scale [20, 10]. Premetric electrodynamics obtains light-cone and metric information from constitutive structure [12, 24], while dispersion-relation and Hamilton-geometric approaches study spacetime structures compatible with matter propagation [23, 2, 16]. Operational detector constructions provide another route to invariant interval information [22].

The present paper addresses a different and more specific inverse problem. We do not derive spacetime dimension, Lorentzian signature, or the relativistic massive shell from generic matter propagation. Instead, the internal relativistic massive-generator geometry is taken from the companion flat-space analysis [9]. The question is whether this internal geometry can be soldered to the tangent bundle of a curved spacetime from a scalar operational incidence response.

The starting variables of the companion theory are the motion–staticity (MS) variables

$$(\mathbf{D}, S), \quad |\mathbf{D}|^2 + S^2 = 1, \quad S > 0. \quad (1)$$

Their operational realization is

$$\mathbf{D} = \frac{\mathbf{v}}{c}, \quad S = \frac{d\tau}{dt} = \sqrt{1 - \frac{|\mathbf{v}|^2}{c^2}}. \quad (2)$$

The physical state manifold is the open upper hemisphere. It is bounded but not compact; adjoining  $S = 0$  gives a compact closure. For  $S > 0$ , the projective map

$$\Phi(\mathbf{D}, S) = \left( \frac{1}{S}, \frac{\mathbf{D}}{S} \right) \quad (3)$$

identifies this hemisphere diffeomorphically with the future unit timelike hyperboloid.

The flat-space construction supplies two further ingredients used here. Its operational realization reconstructs the Minkowski timelike quadratic structure of translation space, while independent rigidity arguments select the standard relativistic free-particle Hamiltonian and massive generator shell. The resulting shell carries the connected Lorentz covariance structure, and the observer-completeness assumptions identify the corresponding connected observer group [9]. The present work takes this relativistic massive-generator geometry as input. In particular, the algebraic relation

$$|\mathbf{D}|^2 + S^2 = 1$$

is not used by itself to derive a curved Lorentzian metric or gravitational dynamics.

The transition to curved spacetime changes the inverse problem in an essential way. The flat construction uses an affine translation space and its canonical dual, whereas a generic curved manifold has no canonical global Abelian translation space. We therefore retain the local Lorentzian generator geometry selected in the flat theory while removing its prior identification with spacetime translations. The missing relation between internal generators and tangent directions is then a soldering problem.

Accordingly, let  $M$  be a smooth four-dimensional manifold and let

$$V \longrightarrow M \quad (4)$$

be an independent rank-four oriented and time-oriented real vector bundle carrying a Lorentzian form

$$\eta : V \otimes V \longrightarrow \mathbb{R} \quad (5)$$

of signature  $(-+++)$ . At the reconstruction stage no spacetime metric, coframe, or affine connection on  $M$ , and no identification

$$TM \simeq V,$$

is assumed. In tetrad and Cartan geometry such an identification is provided by the solder form [18, 26]. Here we ask whether it can instead be reconstructed from scalar incidences between relativistic generators and tangent displacements.

For a massive species of rest mass  $m$ , write

$$Q = mc. \quad (6)$$

We distinguish throughout the Lorentz four-momentum vector  $P \in V_x$  from the generator covector  $\Pi = P^b \in V_x^*$ . With the  $(-+++)$  convention, the positive-energy massive covector shell is

$$\Sigma_{Q,x}^+ = \left\{ \Pi \in V_x^* : \eta_x^*(\Pi, \Pi) = -Q^2, \Pi \text{ on the positive-energy sheet} \right\}. \quad (7)$$

Its positive cone is

$$C_{Q,x} := \text{cone}(\Sigma_{Q,x}^+). \quad (8)$$

This is the positive timelike covector cone together with the origin and is independent of the numerical value of  $Q > 0$ . The fixed shell, rather than the cone, carries the absolute mass normalization.

The additional primitive datum is a scalar response

$$\Theta_x : C_{Q,x} \times T_x M \longrightarrow \mathbb{R}. \quad (9)$$

It is interpreted as an infinitesimal generator–displacement incidence response. No bilinear pairing on  $V_x^* \times T_x M$  and no vector-valued coframe are assumed initially. We require additive composition and positive homogeneity in the generator argument, additive and continuous composition in the displacement argument, smooth base-point dependence, and operational faithfulness.

Since the positive massive cone generates  $V_x^*$ , these assumptions extend the response uniquely to a bilinear pairing

$$\widetilde{\Theta}_x : V_x^* \times T_x M \longrightarrow \mathbb{R}. \quad (10)$$

Finite-dimensional duality then determines a unique linear map

$$e_x : T_x M \longrightarrow V_x, \quad \widetilde{\Theta}_x(\Pi, \xi) = \Pi(e_x \xi). \quad (11)$$

Operational faithfulness is equivalent to

$$\ker e_x = \{0\}. \quad (12)$$

Because  $\dim T_x M = \dim V_x = 4$ , the map is an isomorphism. Under the stated smoothness assumptions the pointwise maps assemble into a smooth solder map

$$e : TM \xrightarrow{\sim} V. \quad (13)$$

For a globally defined response this also entails the topological compatibility condition  $TM \simeq V$ .

The internal Lorentzian form then induces the spacetime metric

$$g = e^* \eta. \quad (14)$$

The dual map transfers the internal fixed- $Q$  shell to the physical cotangent shell

$$\Sigma_{Q,x}^{\text{phys}} = e_x^*(\Sigma_{Q,x}^+),$$

which satisfies

$$g_x^{-1}(p, p) = -Q^2. \quad (15)$$

The normalization of the response is important. If

$$\Theta_x \mapsto \lambda(x)\Theta_x, \quad \lambda(x) > 0,$$

then

$$e_x \mapsto \lambda(x)e_x, \quad g_x \mapsto \lambda^2(x)g_x.$$

Thus an uncalibrated response determines the conformal metric class, whereas an absolutely calibrated response transfers the fixed internal mass scale to an absolute spacetime metric normalization.

The individual mathematical ingredients used in this reconstruction are standard once a solder map is already available. The pairing  $\Pi(e\xi)$  is the usual momentum–coframe incidence, extension from a generating cone is a group-completion construction, and  $V^{**} \simeq V$  is ordinary finite-dimensional duality. The inverse content lies in the primitive formulation: scalar responses are specified initially only on the physically distinguished positive massive-generator sector, and explicit composition, regularity, calibration, and faithfulness assumptions reconstruct the nondegenerate solder map. The local information after extension is equivalent to that of a coframe; the result concerns its operational reconstruction rather than a reduction of its local degrees of freedom.

This distinguishes the present construction from causal, premetric, and dispersion-based reconstruction programs. Those approaches use different physical inputs to recover or constrain conformal or metric spacetime geometry [12, 24, 23]. Here the four-dimensional internal Lorentzian geometry is already supplied by the flat-state rigidity analysis. What is reconstructed is its nondegenerate incidence with an independently given tangent bundle.

Gravitational dynamics constitute a separate, downstream problem. General first-order, Poincaré-gauge, and metric-affine theories allow connection and torsion sectors substantially broader than the Levi–Civita theory [13, 11, 21]. Accordingly, Section 5 considers only a restricted classical branch. After the soldered Lorentzian metric has been reconstructed, we assume an auxiliary torsion-free connection branch, reduce to the metric phase space, impose the standard Lorentzian hypersurface-deformation algebra, and restrict the scalar constraint to

$$\mathcal{H}_\perp = \frac{A}{\sqrt{h}} (\pi^{ij} \pi_{ij} - \alpha \pi^2) - B \sqrt{h} {}^{(3)}R + C \sqrt{h}. \quad (16)$$

Within this specified class, off-shell closure removes the trace-gradient obstruction and gives

$$\alpha = \frac{1}{2}, \quad AB = 1. \quad (17)$$

The resulting constraints are the ADM constraints after parametrizing the remaining overall scale by  $\kappa$  and the ultralocal term by  $\Lambda$ . The standard Legendre transform and Gauss–Codazzi identity then give the Einstein–Hilbert bulk theory. This is a restricted HKT-type completion; it is not a general uniqueness theorem for gravitational dynamics, and neither  $G$  nor  $\Lambda$  is determined.

The logical structure is therefore asymmetric. The principal result is

$$\begin{array}{l} \text{relativistic massive-generator geometry} \\ + \\ \text{composition-compatible, calibrated, faithful} \\ \text{generator–displacement response} \end{array} \longrightarrow \begin{array}{l} e : TM \xrightarrow{\sim} V, \\ g = e^* \eta, \end{array} \quad (18)$$

whereas

$$\begin{array}{l} \text{reconstructed Lorentzian soldering} \\ + \\ \text{restricted canonical assumptions} \end{array} \longrightarrow \text{ADM/Einstein branch} \quad (19)$$

is a conditional downstream completion.

The paper is organized as follows. Section 2 gives the covariant local interpretation of the MS variables and distinguishes their bounded, hyperbolic, vector, and generator-covector representations. Section 3 expresses ordinary Lorentz transport and curvature on the bounded MS fiber. Section 4

proves the operational soldering theorem and the associated metric and massive-shell reconstruction. Section 5 develops the restricted classical gravitational completion, and Section 6 discusses the logical status, limitations, and extensions of the construction. The appendices provide the technical cone-extension and smoothness arguments, the canonical closure calculation, and representative FLRW and Schwarzschild checks.

## 2 Local MS Observer States in Curved Spacetime

The motion–staticity (MS) variables provide a bounded representation of standard massive relativistic kinematics. In the companion flat-space analysis of Ref. [9], the operational identification of the bounded state reconstructs the flat timelike Minkowski structure, while the subsequent rigidity results select the standard relativistic free-particle Hamiltonian and massive generator shell. The observer-completeness assumptions of that analysis then identify the connected Lorentz observer structure. These results are taken as input here. In particular, the pointwise relation

$$|\mathbf{D}|^2 + S^2 = 1, \quad S > 0, \quad (20)$$

is not by itself used to derive Lorentzian spacetime geometry.

We work in the common-observer sector satisfying the inter-species observer assumptions of Ref. [9], for which the conversion scale  $c$  is shared by the massive species under consideration. No identification of this scale with the propagation speed of a particular field is required in the present construction. We use the same metric-signature convention as Ref. [9],

$$\text{sign}(g) = (-, +, +, +). \quad (21)$$

This section proceeds in the forward direction. We provisionally assume a Lorentzian spacetime metric and identify the covariant meaning of the MS variables, their clock interpretation, and their relation to the usual future-timelike hyperboloid. The logical direction will be reversed in Section 4, where the tangent bundle and an internal Lorentz bundle are initially independent and the solder map is reconstructed from an operational incidence response.

Let  $(M, g)$  be an oriented and time-oriented four-dimensional Lorentzian manifold. Let  $u^\mu$  be the future-directed four-velocity of a local observer and  $U^\mu$  the future-directed four-velocity of a massive particle at the same event, normalized by

$$g_{\mu\nu}u^\mu u^\nu = g_{\mu\nu}U^\mu U^\nu = -c^2. \quad (22)$$

### 2.1 Observer decomposition, clock rate, and local evolution

The observer  $u^\mu$  defines the spatial projector

$$h(u)^\mu{}_\nu = \delta^\mu{}_\nu + \frac{1}{c^2}u^\mu u_\nu \quad (23)$$

and the relative Lorentz factor

$$\gamma(U, u) := -\frac{U_\mu u^\mu}{c^2}. \quad (24)$$

The spatial velocity measured by the observer is

$$v^\mu(U, u) := \frac{1}{\gamma(U, u)}h(u)^\mu{}_\nu U^\nu, \quad u_\mu v^\mu = 0. \quad (25)$$

Hence

$$U^\mu = \gamma(U, u) (u^\mu + v^\mu), \quad (26)$$

and normalization of  $U^\mu$  gives

$$\gamma^{-2} = 1 - \frac{v^2}{c^2}, \quad v^2 := g_{\mu\nu} v^\mu v^\nu. \quad (27)$$

Define

$$S(U, u) := \gamma(U, u)^{-1}, \quad D^\mu(U, u) := \frac{v^\mu(U, u)}{c}. \quad (28)$$

Then

$$u_\mu D^\mu = 0, \quad D^2 + S^2 = 1, \quad (29)$$

where

$$D^2 := g_{\mu\nu} D^\mu D^\nu,$$

and conversely

$$U^\mu = \frac{1}{S} (u^\mu + c D^\mu). \quad (30)$$

Thus the MS constraint is a bounded, observer-adapted representation of the standard timelike normalization condition. Curvature does not deform the local state surface. It affects the comparison of states and observer frames at different events.

The scalar  $S$  has a local clock interpretation. Along the particle worldline,

$$dx^\mu = U^\mu d\tau,$$

define the infinitesimal temporal interval assigned by the observer  $u^\mu$  through

$$c^2 d\tau_u := -u_\mu dx^\mu. \quad (31)$$

This is a local observer measurement; no global observer time function is assumed. Equation (24) gives

$$d\tau_u = \gamma(U, u) d\tau, \quad S(U, u) = \frac{d\tau}{d\tau_u}. \quad (32)$$

Accordingly,  $S$  is an observer-relative clock-rate factor. It is neither a curvature scalar nor a gravitational potential. For an observer instantaneously comoving with the particle,

$$D^\mu = 0, \quad S = 1, \quad (33)$$

independently of the local curvature.

For later applications, consider the evolution of  $S$  along the particle worldline. If the particle is freely falling,

$$U^\nu \nabla_\nu U^\mu = 0, \quad (34)$$

then differentiation of Eq. (24) yields

$$\frac{dS}{d\tau} = \frac{S^2}{c^2} U^\mu U^\nu \nabla_\mu u_\nu. \quad (35)$$

Decompose the observer congruence as

$$\nabla_\mu u_\nu = -\frac{1}{c^2}u_\mu a_\nu + \frac{1}{3}\vartheta h(u)_{\mu\nu} + \sigma_{\mu\nu} + \omega_{\mu\nu}, \quad (36)$$

with

$$a_\mu := u^\nu \nabla_\nu u_\mu, \quad \vartheta := \nabla_\mu u^\mu.$$

Using  $U = S^{-1}(u + v)$ , one obtains

$$\frac{dS}{d\tau} = \frac{1}{c^2} \left( a_\mu v^\mu + \frac{\vartheta}{3} v^2 + \sigma_{\mu\nu} v^\mu v^\nu \right). \quad (37)$$

The vorticity drops out because  $v^\mu v^\nu \omega_{\mu\nu} = 0$ . Thus the evolution of  $S$  relative to a prescribed observer field records acceleration, expansion, and shear of that congruence; it is not an invariant measure of curvature.

For a nongeodesic particle with four-acceleration

$$A^\mu := U^\nu \nabla_\nu U^\mu, \quad (38)$$

the corresponding exact relation is

$$\frac{dS}{d\tau} = \frac{S^2}{c^2} (U^\mu U^\nu \nabla_\mu u_\nu + u_\mu A^\mu). \quad (39)$$

A useful specialization is the 3 + 1 decomposition. Let  $\Sigma_t$  be a foliation by spacelike hypersurfaces and let  $n^\mu$  denote the future-directed normal observer four-velocity,

$$n_\mu n^\mu = -c^2. \quad (40)$$

Thus  $n^\mu$  is a four-velocity rather than the dimensionless unit normal; the latter is  $\widehat{n}^\mu = n^\mu/c$ . Write the evolution vector as

$$\left( \frac{\partial}{\partial t} \right)^\mu = N n^\mu + \beta^\mu, \quad n_\mu \beta^\mu = 0. \quad (41)$$

With this convention  $N$  converts coordinate time into the proper-time direction of the normal observers. For a particle trajectory parametrized by  $t$ ,

$$\gamma(U, n) = N \frac{dt}{d\tau}, \quad (42)$$

and therefore

$$\frac{d\tau}{dt} = NS(U, n). \quad (43)$$

The factors  $N$  and  $S$  have distinct roles:  $S$  is the local special-relativistic clock factor relative to the normal observer, whereas  $N$  belongs to the chosen spacetime foliation. In particular, for a particle momentarily comoving with that observer,

$$S = 1, \quad \frac{d\tau}{dt} = N.$$

## 2.2 Bounded, hyperbolic, and generator representations

At a fixed event  $x$ , relative to a fixed observer  $u$ , choose an orthonormal basis of the Euclidean rest space  $u^\perp$ . The massive MS state space is represented by

$$\mathcal{S}_{x,u} = \{(\mathbf{D}, S) : |\mathbf{D}|^2 + S^2 = 1, S > 0\}. \quad (44)$$

It is bounded but not compact. Its closure, obtained by adjoining  $S = 0$ , is compact. Whenever the phrase “compact MS representation” is used below, it refers to this bounded compactification and not to compactness of the physical massive  $S > 0$  state manifold.

Define

$$X^A = \Phi(\mathbf{D}, S) := \left(\frac{1}{S}, \frac{\mathbf{D}}{S}\right). \quad (45)$$

With

$$\eta_{AB} = \text{diag}(-1, 1, 1, 1), \quad (46)$$

one has

$$\eta_{AB} X^A X^B = -1, \quad X^0 > 0. \quad (47)$$

Hence  $\Phi$  is a diffeomorphism from the open upper hemisphere onto the future unit timelike hyperboloid.

**Proposition 2.1** (Fiberwise equivalence). *For fixed  $x \in M$  and a future-directed observer four-velocity  $u \in T_x M$ , the map*

$$(D, S) \mapsto U = S^{-1}(u + cD) \quad (48)$$

*is a bijection between  $\mathcal{S}_{x,u}$  and*

$$\mathcal{H}_x^+ = \{U \in T_x M : g(U, U) = -c^2, U \text{ future directed}\}. \quad (49)$$

*Proof.* Since  $D \in u^\perp$ ,

$$\begin{aligned} g(U, U) &= \frac{1}{S^2} [g(u, u) + c^2 g(D, D)] \\ &= \frac{c^2}{S^2} (-1 + D^2) = -c^2, \end{aligned} \quad (50)$$

where  $D^2 = 1 - S^2$ . Since  $S > 0$ , the future time orientation is preserved. Conversely, Eqs. (24) and (28) recover  $S$  and  $D$  uniquely from every  $U \in \mathcal{H}_x^+$ .  $\square$

This is an equivalence of representations rather than a second massive kinematics. The hyperboloid variables make the linear Lorentz action manifest; the MS variables provide bounded coordinates and isolate the observer-relative clock factor.

The two descriptions carry naturally related Riemannian metrics on the state fiber. The metric induced from the round metric on the unit three-sphere is

$$g_{\text{round}} = d\mathbf{D}^2 + \frac{(\mathbf{D} \cdot d\mathbf{D})^2}{1 - |\mathbf{D}|^2}. \quad (51)$$

The standard hyperbolic metric on the future unit hyperboloid, pulled back through  $\Phi$ , is

$$g_H = \frac{d\mathbf{D}^2}{1 - |\mathbf{D}|^2} + \frac{(\mathbf{D} \cdot d\mathbf{D})^2}{(1 - |\mathbf{D}|^2)^2}. \quad (52)$$

Since  $S^2 = 1 - |\mathbf{D}|^2$ ,

$$g_H = S^{-2} g_{\text{round}}. \quad (53)$$

The boundary  $S \rightarrow 0^+$  is therefore at finite distance in the auxiliary round metric but at infinite distance in the hyperbolic metric. For a fixed massive shell it is the infinite-rapidity boundary.

We now distinguish the Lorentz vector representing four-momentum from the covector that pairs with translations or displacements. With the massive normalization inherited from Ref. [9],

$$Q = mc, \quad (54)$$

define the future-directed Lorentz vector

$$P^A := QX^A = Q\left(\frac{1}{S}, \frac{\mathbf{D}}{S}\right). \quad (55)$$

Thus

$$P^A = \left(\frac{E}{c}, \mathbf{p}\right), \quad E = \frac{mc^2}{S}, \quad \mathbf{p} = Q\frac{\mathbf{D}}{S} = mc\frac{\mathbf{D}}{S}, \quad (56)$$

and

$$\eta_{AB}P^AP^B = -Q^2. \quad (57)$$

The corresponding Lorentz covector is

$$\Pi_A := \eta_{AB}P^B = \left(-\frac{E}{c}, \mathbf{p}\right). \quad (58)$$

It obeys

$$\eta^{AB}\Pi_A\Pi_B = -Q^2. \quad (59)$$

With the present  $(- + ++)$  convention, positive energy is equivalently the condition

$$-\Pi_{\hat{0}} = \frac{E}{c} > 0. \quad (60)$$

We maintain this distinction below:  $P^A$  denotes the positive-energy Lorentz vector, whereas  $\Pi_A$  denotes its covector form. In the flat theory  $\Pi$  is the canonical generator that pairs with spacetime displacement; in the curved reconstruction it is this covector object that will be fibered in the internal dual bundle and used in the response  $\Theta_x(\Pi, \xi)$ .

Although  $\mathbf{D}$  is bounded,

$$|\mathbf{p}| = Q\frac{|\mathbf{D}|}{S} \quad (61)$$

is unbounded as  $S \rightarrow 0^+$ . The bounded MS representation therefore introduces neither a maximal momentum nor an ultraviolet cutoff.

The boundary  $S = 0$  must also be distinguished from a physical massive state becoming null. At fixed  $Q > 0$ ,

$$S \rightarrow 0^+ \quad (62)$$

is an infinite-rapidity and infinite-energy limit on the massive shell. The companion flat-space theory admits a separate scale-free null completion, obtainable through the finite-energy  $Q \rightarrow 0^+$  limit or a homogeneous lift of the projective boundary. The present paper does not develop the corresponding curved-spacetime null soldering sector. Accordingly, adjoining  $S = 0$  to the bounded massive coordinates is not by itself a construction of general massless dynamics.

Once a Lorentzian spacetime metric is given, the future-timelike observer fibers form the standard observer bundle. For an oriented and time-oriented spacetime,

$$O = P_{\text{SO}^+(1,3)}(M) \times_{\text{SO}^+(1,3)} \mathbb{H}^3. \quad (63)$$

Let  $\mathbb{S}_+^3$  denote the open upper hemisphere above. Through  $\Phi$ , the same bundle can be represented as

$$\mathcal{S}_{\text{MS}} = P_{\text{SO}^+(1,3)}(M) \times_{\text{SO}^+(1,3)} \mathbb{S}_+^3, \quad (64)$$

where the Lorentz action on the bounded coordinates is

$$T_\Lambda = \Phi^{-1} \circ \Lambda \circ \Phi. \quad (65)$$

Thus

$$\mathcal{S}_{\text{MS}} \simeq O$$

fiberwise. The MS variables are nonlinear bounded coordinates on the ordinary timelike observer fiber; they do not define an independent observer geometry.

The equation

$$D^2 + S^2 = 1$$

is therefore entirely local. It contains no prescription for comparing states at distinct spacetime events and does not by itself identify an internal Lorentz space with  $T_x M$ . Once a soldered Lorentzian geometry is available, inter-event comparison is governed by the ordinary Lorentz connection, whose action in the bounded variables is described in Section 3. Section 4 then addresses the logically separate inverse problem: starting from an abstract spacetime manifold, an independent internal Lorentz bundle, and a scalar generator–displacement response, it reconstructs the solder map without assuming a spacetime metric or coframe at the outset.

### 3 Lorentz Transport on the Bounded MS State Fiber

Section 2 identified the MS variables as bounded coordinates on the ordinary future-timelike observer fiber. We now describe how the standard Lorentz connection acts in these coordinates.

Throughout this section the Lorentzian metric, orthonormal frame bundle, and metric-compatible Lorentz connection are regarded as given. No new connection or curvature tensor is introduced. The construction is the push-forward of ordinary Lorentz transport to the bounded MS representation and therefore presupposes both the metric and the solder form. The inverse reconstruction begins only in Section 4.

#### 3.1 Observer-adapted transport and the nonlinear Lorentz action

Let

$$e_{\hat{a}}^\mu, \quad \hat{a} = 0, 1, 2, 3, \quad (66)$$

be a local oriented and time-oriented orthonormal tetrad,

$$g_{\mu\nu} e_{\hat{a}}^\mu e_{\hat{b}}^\nu = \eta_{\hat{a}\hat{b}}, \quad \eta_{\hat{a}\hat{b}} = \text{diag}(-1, 1, 1, 1). \quad (67)$$

We adapt its timelike leg to the chosen observer,

$$e_{\hat{0}}^\mu = \frac{u^\mu}{c}. \quad (68)$$

For a massive four-velocity  $U^\mu$ , define the dimensionless tetrad components

$$X^{\hat{a}} := \frac{1}{c} e^{\hat{a}}_\mu U^\mu. \quad (69)$$

In the observer-adapted frame,

$$X^{\hat{a}} = \frac{1}{S} (1, D^i), \quad \delta_{ij} D^i D^j + S^2 = 1. \quad (70)$$

Thus  $X^{\hat{a}}$  is the usual future unit timelike vector,

$$\eta_{\hat{a}\hat{b}} X^{\hat{a}} X^{\hat{b}} = -1, \quad X^{\hat{0}} > 0,$$

written in the projective coordinates of Section 2.

Let

$$\omega^{\hat{a}}_{\hat{b}}$$

be a metric-compatible Lorentz connection,

$$\omega_{\hat{a}\hat{b}} = -\omega_{\hat{b}\hat{a}}.$$

Relative to the observer-adapted splitting, write

$$\kappa^i := \omega^{\hat{i}}_{\hat{0}}, \quad \varpi^i_j := \omega^{\hat{i}}_{\hat{j}}, \quad \varpi_{ij} = -\varpi_{ji}. \quad (71)$$

The one-forms  $\varpi^i_j$  describe spatial frame rotation, while  $\kappa^i$  is the boost part of the same Lorentz connection.

Under a change of local Lorentz frame

$$\Lambda(x) \in SO^+(1, 3),$$

we use

$$X \mapsto \Lambda X, \quad \omega \mapsto \Lambda \omega \Lambda^{-1} - d\Lambda \Lambda^{-1}. \quad (72)$$

The induced transformation of the bounded variables is therefore

$$(\mathbf{D}, S) \mapsto \Phi^{-1} \circ \Lambda \circ \Phi(\mathbf{D}, S). \quad (73)$$

A Lorentz transformation acts linearly on  $X$  and nonlinearly on  $(\mathbf{D}, S)$ ; the two descriptions represent the same underlying timelike state.

Suppose that  $X^{\hat{a}}$  is parallel transported along a curve:

$$DX^{\hat{a}} := dX^{\hat{a}} + \omega^{\hat{a}}_{\hat{b}} X^{\hat{b}} = 0. \quad (74)$$

For the tetrad components of a freely falling four-velocity this is the tetrad form of the geodesic equation.

**Proposition 3.1** (Lorentz transport in MS variables). *Parallel transport of  $X = S^{-1}(1, \mathbf{D})$  is equivalent to*

$$dS = S \kappa_i D^i, \quad (75)$$

$$dD^i = -\varpi^i_j D^j - \kappa^i + (\kappa_j D^j) D^i. \quad (76)$$

The constraint

$$D^2 + S^2 = 1 \quad (77)$$

is invariant under this transport.

*Proof.* The temporal component of Eq. (74) is

$$d(S^{-1}) + \kappa_i \frac{D^i}{S} = 0, \quad (78)$$

and gives Eq. (75). The spatial components are

$$d\left(\frac{D^i}{S}\right) + \kappa^i \frac{1}{S} + \varpi^i_j \frac{D^j}{S} = 0, \quad (79)$$

which, together with Eq. (75), gives Eq. (76).

Finally,

$$\begin{aligned} d(D^2 + S^2) &= 2D_i dD^i + 2S dS \\ &= 2(\kappa_i D^i)(D^2 + S^2 - 1), \end{aligned} \quad (80)$$

where antisymmetry of  $\varpi_{ij}$  has been used. Hence the constraint surface is preserved.  $\square$

A pure spatial rotation changes only the direction of  $\mathbf{D}$ . The boost part changes both  $S$  and  $\mathbf{D}$ . The quadratic term in Eq. (76) is the projective correction required to keep the state on the bounded hemisphere.

The infinitesimal Lorentz action can be written directly as vector fields on the MS state manifold. Define

$$\mathbf{B}_i = \left(\delta_i^j - D_i D^j\right) \frac{\partial}{\partial D^j} - S D_i \frac{\partial}{\partial S}, \quad (81)$$

$$\mathbf{J}_{ij} = D_j \frac{\partial}{\partial D^i} - D_i \frac{\partial}{\partial D^j}. \quad (82)$$

Both are tangent to  $D^2 + S^2 = 1$ . They are the fundamental vector fields associated with the left projective action

$$T_\Lambda = \Phi^{-1} \circ \Lambda \circ \Phi.$$

With the standard Lie bracket of vector fields,

$$[\mathbf{B}_i, \mathbf{B}_j] = -\mathbf{J}_{ij}, \quad (83)$$

$$[\mathbf{J}_{ij}, \mathbf{B}_k] = \delta_{ik} \mathbf{B}_j - \delta_{jk} \mathbf{B}_i, \quad (84)$$

and

$$[\mathbf{J}_{ij}, \mathbf{J}_{kl}] = -\delta_{jk}\mathbf{J}_{il} + \delta_{ik}\mathbf{J}_{jl} + \delta_{jl}\mathbf{J}_{ik} - \delta_{il}\mathbf{J}_{jk}. \quad (85)$$

The signs should not be confused with those of a matrix representation of the Lorentz algebra. For the convention used here, fundamental vector fields generated by a left group action form an anti-homomorphism:

$$[A^\#, B^\#] = -[A, B]^\#.$$

Equations (83)–(85) are the corresponding vector-field realization on the bounded state manifold.

The transport equations may be summarized by the horizontal operator

$$\mathcal{D} = d + \kappa^i \mathbf{B}_i + \frac{1}{2} \varpi^{ij} \mathbf{J}_{ij}, \quad (86)$$

with

$$\mathcal{D}S = 0, \quad \mathcal{D}D^i = 0$$

understood as shorthand for Eqs. (75) and (76). Because the vector fields above realize a left action anti-homomorphically,  $\mathcal{D}$  is best regarded as the nonlinear horizontal transport induced by the ordinary Lorentz connection. It is not an additional matrix-valued gauge connection; in particular, its curvature should not be computed by treating Eq. (86) as though  $\mathbf{B}_i, \mathbf{J}_{ij}$  formed a homomorphic matrix representation.

The distinction introduced in Section 2 between four-momentum vectors and generator covectors is compatible with this transport. For fixed  $Q$ ,

$$P^{\hat{a}} = QX^{\hat{a}} \quad (87)$$

obeys the same vector transport law,

$$DP^{\hat{a}} = 0.$$

Its metric-dual covector

$$\Pi_{\hat{a}} = \eta_{\hat{a}\hat{b}} P^{\hat{b}}$$

is transported by the dual connection,

$$D\Pi_{\hat{a}} := d\Pi_{\hat{a}} - \omega^{\hat{b}}_{\hat{a}} \Pi_{\hat{b}} = 0. \quad (88)$$

Section 4 will use this covector object as the generator argument of the operational response.

### 3.2 Curvature, holonomy, and inter-event comparison

The Lorentz curvature two-form is

$$R^{\hat{a}}_{\hat{b}} = d\omega^{\hat{a}}_{\hat{b}} + \omega^{\hat{a}}_{\hat{c}} \wedge \omega^{\hat{c}}_{\hat{b}}. \quad (89)$$

For an internal Lorentz vector,

$$D^2 X^{\hat{a}} = R^{\hat{a}}_{\hat{b}} X^{\hat{b}}. \quad (90)$$

Relative to the observer splitting,

$$\begin{aligned}\mathcal{K}^i &:= R^i_{\hat{0}} = d\kappa^i + \varpi^i_j \wedge \kappa^j, \\ \mathcal{R}^i_j &:= R^i_{\hat{j}} = d\varpi^i_j + \varpi^i_k \wedge \varpi^k_j + \kappa^i \wedge \kappa_j.\end{aligned}\tag{91}$$

The second term is the rotational curvature, while  $\mathcal{K}^i$  is the boost component of the same Lorentz curvature; they are not independent field strengths.

Pushing the infinitesimal Lorentz action to the bounded state fiber gives the curvature vector field

$$\mathcal{R} = \mathcal{K}^i \mathbf{B}_i + \frac{1}{2} \mathcal{R}^{ij} \mathbf{J}_{ij}.\tag{92}$$

This equation is a push-forward of Eq. (89), rather than the definition of a new curvature.

For a sufficiently small oriented loop with area bivector  $\Sigma^{\mu\nu}$ , our parallel-transport convention  $DX = 0$  gives

$$X \mapsto \left[ I - \frac{1}{2} R_{\mu\nu} \Sigma^{\mu\nu} + O(\Sigma^2) \right] X.\tag{93}$$

The corresponding variation of the bounded variables is

$$\delta(\mathbf{D}, S) = -\frac{1}{2} \mathcal{R}_{\mu\nu} \Sigma^{\mu\nu} + O(\Sigma^2).\tag{94}$$

Reversing the orientation of the loop reverses the sign. Curvature therefore controls infinitesimal and contractible-loop holonomy. Vanishing local curvature does not, on a nonsimply connected region, exclude nontrivial global holonomy.

This distinction is useful for separating two different notions of observer dependence. At a single event, two oriented and time-oriented orthonormal frames are related by an element of  $SO^+(1, 3)$ , and the same timelike state is represented in the two frames by MS coordinates related through

$$T_\Lambda = \Phi^{-1} \Lambda \Phi.$$

This is ordinary local Lorentz covariance. Comparing states or frames at different events requires a transport prescription, and the connection supplies precisely that additional information.

In particular, the scalar  $S$  remains observer dependent even in a curved spacetime. Its value at an event is determined by the relative Lorentz factor,

$$S(U, u) = \left( -\frac{U \cdot u}{c^2} \right)^{-1},$$

not by curvature alone. The curvature controls how observer frames can be compared around loops, while the evolution equation (37) describes how  $S$  changes along a worldline relative to a specified congruence.

The same separation appears in a 3 + 1 description. For the normal observer  $n^\mu$ ,

$$\frac{d\tau}{dt} = NS(U, n),\tag{95}$$

where  $S$  measures local relative motion and  $N$  is the lapse of the chosen foliation. A vanishing or small lapse is therefore not, by itself, a statement about the local MS state, and a small value of  $S$  is not a curvature invariant.

The logical direction of this section is therefore

$$(g, \omega) \longrightarrow \text{Lorentz transport on the bounded MS observer fiber.} \quad (96)$$

No converse reconstruction has been used. The next section removes the assumed soldering relation between internal and tangent directions and asks instead whether a scalar response

$$\Theta_x(\Pi, \xi)$$

with  $\Pi \in V_x^*$  and  $\xi \in T_x M$  is sufficient to recover that relation operationally.

## 4 Operational Soldering and Reconstruction of Lorentzian Geometry

Sections 2 and 3 proceeded in the forward direction: a Lorentzian spacetime geometry was assumed and the MS variables were interpreted as bounded coordinates on its standard timelike observer fibers. We now reverse the part of that construction that identifies internal relativistic generators with spacetime directions.

Let  $M$  be a smooth four-dimensional manifold. At this stage no spacetime metric, coframe, affine connection, orientation, or time orientation on  $M$  is assumed. Let

$$V \longrightarrow M \quad (97)$$

be an independent rank-four real vector bundle equipped with a smooth Lorentzian form

$$\eta_x : V_x \times V_x \longrightarrow \mathbb{R} \quad (98)$$

of signature  $(-, +, +, +)$ , together with an internal orientation and time orientation.

The internal massive-generator geometry is inherited from the standard relativistic branch selected by the rigidity results of Ref. [9]. The present section does not attempt to derive that Lorentzian structure from  $D^2 + S^2 = 1$  alone. For a massive species of rest mass  $m$ , let

$$Q = mc. \quad (99)$$

Following the vector–covector distinction introduced in Section 2, the internal four-momentum vector is denoted by  $P \in V_x$ , whereas the generator that pairs with a displacement is its Lorentz-dual covector

$$\Pi = P^\flat \in V_x^*.$$

The positive-energy internal massive shell is therefore

$$\Sigma_{Q,x}^+ := \left\{ \Pi \in V_x^* : \eta_x^*(\Pi, \Pi) = -Q^2, \Pi \text{ on the positive-energy sheet} \right\}. \quad (100)$$

In an oriented and time-oriented orthonormal internal frame,

$$\eta^* = \text{diag}(-1, 1, 1, 1),$$

this sheet may be parametrized as

$$\Pi(\mathbf{p}) = \left( -\sqrt{Q^2 + |\mathbf{p}|^2}, \mathbf{p} \right), \quad (101)$$

so that

$$E = -c \Pi_{\hat{0}} > 0.$$

This sign convention is the covector counterpart of  $P^{\hat{a}} = (E/c, \mathbf{p})$ .

The issue is not the existence of this internal Lorentzian generator geometry, but its relation to tangent displacement. The data  $(M, V, \eta)$  do not provide such a relation. If one isomorphism

$$e_x : T_x M \longrightarrow V_x$$

were already available, then for every  $B_x \in GL(T_x M)$  the map

$$e'_x = e_x \circ B_x$$

would be another isomorphism into the same internal Lorentz space, while in general

$$(e'_x)^* \eta_x = B_x^*(e_x^* \eta_x) \neq e_x^* \eta_x.$$

An additional incidence datum is therefore required.

Once a coframe is given, the scalar pairing

$$\Pi(e_x \xi)$$

is standard in tetrad and first-order descriptions of relativistic matter [13, 21]. The question addressed below is the inverse one: whether the solder map can be reconstructed when only a scalar generator–displacement response is specified operationally on the positive massive sector.

## 4.1 Generator–displacement response and operational soldering

We first record the algebraic property of the positive massive shell needed for the reconstruction.

**Lemma 4.1** (Spanning property of the positive massive shell). *For every  $x \in M$  and every  $Q > 0$ ,*

$$\text{span } \Sigma_{Q,x}^+ = V_x^*. \quad (102)$$

*Proof.* Choose an internal orthonormal dual basis  $\{\varepsilon^0, \varepsilon^1, \varepsilon^2, \varepsilon^3\}$  with the positive-energy direction chosen so that the rest covector is

$$\Pi_0 = -Q \varepsilon^0. \quad (103)$$

For any  $a > 0$ , the shell also contains

$$\Pi_i^\pm = -\sqrt{Q^2 + a^2} \varepsilon^0 \pm a \varepsilon^i, \quad i = 1, 2, 3. \quad (104)$$

Their differences satisfy

$$\Pi_i^+ - \Pi_i^- = 2a \varepsilon^i. \quad (105)$$

Thus the three spatial basis covectors lie in the span of the shell, while  $\Pi_0$  supplies the timelike direction.  $\square$

Let

$$C_{Q,x} := \text{cone}(\Sigma_{Q,x}^+) \subset V_x^* \quad (106)$$

be the cone of finite positive combinations of shell covectors. It has the intrinsic description

$$C_{Q,x} = \{0\} \cup \{\Pi \in V_x^* : \eta_x^*(\Pi, \Pi) < 0, \Pi \text{ on the positive-energy side}\}. \quad (107)$$

Although the notation retains  $Q$ , the set on the right-hand side is independent of  $Q > 0$ .

Indeed, for any nonzero positive-energy timelike covector  $\Pi$ , set

$$r(\Pi) := \sqrt{-\eta_x^*(\Pi, \Pi)}.$$

Then

$$\widehat{\Pi} = \frac{Q}{r(\Pi)} \Pi \in \Sigma_{Q,x}^+, \quad \Pi = \frac{r(\Pi)}{Q} \widehat{\Pi}. \quad (108)$$

Conversely, the positive-energy timelike cone is convex, so positive sums remain in the same cone. In particular,

$$C_{Q,x} - C_{Q,x} = V_x^*. \quad (109)$$

The cone will be used as an additive generator domain. Elements away from the fixed shell need not be interpreted as one-particle states of rest mass  $Q/c$ .

The primitive incidence datum is a real scalar response

$$\Theta_x : C_{Q,x} \times T_x M \longrightarrow \mathbb{R}. \quad (110)$$

It is interpreted as an infinitesimal generator–displacement response. At this point neither a bilinear pairing on  $V_x^* \times T_x M$  nor a vector-valued coframe is assumed.

We impose the following conditions.

**Assumption 4.2** (Generator composition). *For fixed  $\xi \in T_x M$  and  $\Pi_1, \Pi_2 \in C_{Q,x}$ ,*

$$\Theta_x(\Pi_1 + \Pi_2, \xi) = \Theta_x(\Pi_1, \xi) + \Theta_x(\Pi_2, \xi), \quad (111)$$

$$\Theta_x(\lambda \Pi, \xi) = \lambda \Theta_x(\Pi, \xi), \quad \lambda \geq 0. \quad (112)$$

**Assumption 4.3** (Displacement composition). *For fixed  $\Pi \in C_{Q,x}$ ,*

$$\Theta_x(\Pi, \xi + \zeta) = \Theta_x(\Pi, \xi) + \Theta_x(\Pi, \zeta), \quad (113)$$

*and  $\xi \mapsto \Theta_x(\Pi, \xi)$  is continuous.*

**Assumption 4.4** (Operational faithfulness). *If*

$$\Theta_x(\Pi, \xi) = 0 \quad \text{for every } \Pi \in \Sigma_{Q,x}^+, \quad (114)$$

*then  $\xi = 0$ .*

Finally, we assume smooth dependence on the base point in the following sense: if  $\Pi : U \rightarrow C_Q$  is a smooth local generator section and  $\xi \in \Gamma(TU)$  is a smooth vector field, then

$$x \longmapsto \Theta_x(\Pi(x), \xi(x))$$

is smooth.

The operational content of these assumptions may be stated more explicitly. Elements of the fixed shell  $\Sigma_{Q,x}^+$  represent calibrated massive generator probes, whereas the larger cone  $C_{Q,x}$  serves only as the domain for finite positive compositions of such generator data; its generic elements are not interpreted as additional one-particle states of fixed mass. Generator additivity expresses the composition of independent generator contributions to the same infinitesimal displacement, while displacement additivity expresses infinitesimal concatenation at a fixed base point. Operational faithfulness requires that every nonzero  $\xi \in T_x M$  be detected by at least one physical massive generator:

$$\xi \neq 0 \implies \exists \Pi \in \Sigma_{Q,x}^+ \text{ such that } \Theta_x(\Pi, \xi) \neq 0.$$

The cone extension used below therefore parallels the composition logic of the unnormalized generator data in Ref. [9]; it does not enlarge the physical fixed-mass one-particle state space. Positive homogeneity in Eq. (112) is an independent assumption; generator additivity alone is not used to infer arbitrary real homogeneity.

**Lemma 4.5** (Bilinear extension). *Under Assumptions 4.2 and 4.3, the response has a unique bilinear extension*

$$\widetilde{\Theta}_x : V_x^* \times T_x M \longrightarrow \mathbb{R} \quad (115)$$

whose restriction to  $C_{Q,x} \times T_x M$  is the original response.

*Proof.* Equation (109) allows every  $\Pi \in V_x^*$  to be written as  $\Pi = \Pi_+ - \Pi_-$ ,  $\Pi_{\pm} \in C_{Q,x}$ . Define

$$\widetilde{\Theta}_x(\Pi, \xi) := \Theta_x(\Pi_+, \xi) - \Theta_x(\Pi_-, \xi). \quad (116)$$

If  $\Pi = \Pi'_+ - \Pi'_-$  is a second decomposition, then  $\Pi_+ + \Pi'_- = \Pi'_+ + \Pi_-$ . Generator additivity gives  $\Theta_x(\Pi_+, \xi) + \Theta_x(\Pi'_-, \xi) = \Theta_x(\Pi'_+, \xi) + \Theta_x(\Pi_-, \xi)$ , so Eq. (116) is independent of the chosen decomposition.

Additivity and positive homogeneity on the generating cone imply real linearity in the extended generator argument. For fixed  $\Pi$ , additivity and continuity in  $\xi$  imply, by the continuous Cauchy equation,

$$\widetilde{\Theta}_x(\Pi, \lambda \xi) = \lambda \widetilde{\Theta}_x(\Pi, \xi), \quad \lambda \in \mathbb{R}. \quad (117)$$

Hence the extension is bilinear. Uniqueness follows because  $C_{Q,x}$  generates  $V_x^*$ .  $\square$

The extension step is a standard group-completion argument for a generating cone. The reconstruction content lies in the fact that the primitive datum is specified only as a scalar response on the positive massive sector.

**Theorem 4.6** (Operational Soldering Theorem). *Let  $M$  be a smooth four-dimensional manifold and let  $V \rightarrow M$  be a rank-four oriented and time-oriented Lorentz vector bundle. Suppose a globally defined smooth response*

$$\Theta_x : C_{Q,x} \times T_x M \longrightarrow \mathbb{R} \quad (118)$$

*satisfies generator composition, displacement composition, and operational faithfulness.*

*Then there exists a unique smooth bundle isomorphism*

$$e : TM \xrightarrow{\sim} V \quad (119)$$

*such that the bilinear extension of the response satisfies*

$$\widetilde{\Theta}_x(\Pi, \xi) = \Pi(e_x \xi) \quad (120)$$

*for every  $x \in M$ ,  $\Pi \in V_x^*$ , and  $\xi \in T_x M$ .*

*Proof.* By Lemma 4.5, for fixed  $\xi \in T_x M$  the map

$$\Pi \mapsto \widetilde{\Theta}_x(\Pi, \xi)$$

is an element of  $V_x^{**}$ . The canonical finite-dimensional identification  $V_x^{**} \simeq V_x$  therefore gives a unique vector  $e_x \xi \in V_x$  such that

$$\Pi(e_x \xi) = \widetilde{\Theta}_x(\Pi, \xi)$$

for all  $\Pi \in V_x^*$ . Linearity of the extended response in its second argument makes  $e_x : T_x M \longrightarrow V_x$  linear.

If  $e_x \xi = 0$ , then

$$\Theta_x(\Pi, \xi) = 0$$

for every  $\Pi \in \Sigma_{Q,x}^+$ . Operational faithfulness therefore implies  $\xi = 0$ , so

$$\ker e_x = \{0\}. \quad (121)$$

Since both vector spaces have dimension four,  $e_x$  is an isomorphism.

To establish smoothness, choose on a sufficiently small neighborhood  $U \subset M$  four smooth shell sections

$$\Pi^{(A)}, \quad A = 0, 1, 2, 3,$$

whose values form a basis of  $V^*$ , and let  $E_A$  denote the dual local frame of  $V$ . Then

$$e(\xi) = \sum_{A=0}^3 \Theta(\Pi^{(A)}, \xi) E_A. \quad (122)$$

Each coefficient is smooth by assumption, and hence  $e$  is smooth. Uniqueness follows because  $V_x^*$  separates vectors in  $V_x$ .  $\square$

The construction of the theorem is local. The global conclusion follows only when the response is globally defined with the stated smoothness and faithfulness properties. Its existence then entails

$$TM \simeq V, \quad (123)$$

which is a genuine topological compatibility condition and is not asserted for an arbitrary pair  $(M, V)$ .

The role of operational faithfulness is equally precise. After the bilinear extension has been obtained,

$$\widetilde{\Theta}_x(\Pi, \xi) = \Pi(e_x \xi), \quad \Pi \in V_x^*. \quad (124)$$

Restricting this relation back to the physical massive shell gives

$$\Theta_x(\Pi, \xi) = 0 \quad \text{for every } \Pi \in \Sigma_{Q,x}^+ \iff e_x \xi = 0.$$

Thus, in the equal-rank setting considered here, operational faithfulness is the operational form of solder-map nondegeneracy. The theorem does not derive nondegeneracy without the corresponding physical assumption.

Nor does the theorem reduce the number of local degrees of freedom required to specify a solder map. Once the response has been extended bilinearly, it contains equivalent local linear information.

The distinction is one of primitive formulation: the input consists of scalar incidences between physically admissible massive generators and tangent displacements, rather than a vector-valued coframe supplied at the outset.

The reconstruction is finite at each event. If

$$\Pi^{(A)} \in \Sigma_{Q,x}^+, \quad A = 0, 1, 2, 3, \quad (125)$$

form a basis of  $V_x^*$ , define

$$\theta^{(A)}(\xi) := \Theta_x(\Pi^{(A)}, \xi). \quad (126)$$

The four one-forms  $\theta^{(A)}$  form a coframe, and if  $E_A$  is the dual basis of  $V_x$ ,

$$e_x = \sum_{A=0}^3 E_A \otimes \theta^{(A)}. \quad (127)$$

Equation (127) is a local tomography formula for the reconstructed map, not an additional postulate.

## 4.2 Metric and massive-shell reconstruction

The internal Lorentzian form may now be transferred to tangent space.

**Corollary 4.7** (Lorentzian metric reconstruction). *Under the hypotheses of Theorem 4.6,*

$$g_x(\xi, \zeta) := \eta_x(e_x \xi, e_x \zeta) \quad (128)$$

*defines a unique smooth Lorentzian metric on  $M$ .*

*Proof.* Since  $e_x$  is an isomorphism and  $\eta_x$  is nondegenerate with Lorentzian signature, the pullback  $g = e^* \eta$  is a nondegenerate Lorentzian metric. Smoothness follows from that of  $e$  and  $\eta$ .  $\square$

The internal orientation and time orientation may likewise be pulled back through  $e$ , thereby supplying the corresponding structures on  $M$ . In local frames,

$$g_{\mu\nu} = \eta_{ab} e^a{}_\mu e^b{}_\nu. \quad (129)$$

This is the standard tetrad relation. Its role here is reconstructive: the coframe represents the incidence map obtained from the response rather than being the primitive datum from which the response is defined.

The dual formulation is particularly natural for the massive generators. For  $\Pi \in V_x^*$ , define the corresponding spacetime covector

$$p_\Pi \in T_x^* M \quad (130)$$

by

$$p_\Pi(\xi) := \widetilde{\Theta}_x(\Pi, \xi). \quad (131)$$

Theorem 4.6 gives

$$p_\Pi = e_x^* \Pi. \quad (132)$$

Since  $e_x$  is an isomorphism,

$$e_x^* : V_x^* \xrightarrow{\sim} T_x^* M \quad (133)$$

is an isomorphism as well.

The internal shell therefore induces the physical cotangent mass shell

$$\Sigma_{Q,x}^{\text{phys}} := e_x^*(\Sigma_{Q,x}^+) \subset T_x^*M. \quad (134)$$

For every  $p \in \Sigma_{Q,x}^{\text{phys}}$ ,

$$g_x^{-1}(p, p) = -Q^2. \quad (135)$$

The cone and the fixed shell carry different information. The image

$$e_x^*(C_{Q,x}) \quad (136)$$

is the positive timelike cotangent cone. Its boundary is the null cone and hence determines the point-wise conformal Lorentzian structure. The numerical value of  $Q$  does not affect this cone. By contrast, once the generator–displacement response is absolutely calibrated, the image of the fixed- $Q$  shell supplies an absolute quadratic normalization.

For completeness, the following standard quadratic fact makes this statement explicit.

**Corollary 4.8** (Determination of the co-metric by a massive shell). *At fixed  $x$  and fixed  $Q > 0$ , a complete calibrated positive-energy massive shell  $\Sigma_{Q,x}^{\text{phys}}$  uniquely determines the Lorentzian quadratic co-metric  $g_x^{-1}$ .*

*Proof.* Suppose two quadratic forms  $q_1$  and  $q_2$  have the same positive-energy  $Q$ -mass shell. For every shell point  $p$  and every  $\lambda > 0$ ,

$$q_1(\lambda p) = q_2(\lambda p) = -\lambda^2 Q^2. \quad (137)$$

The positive rays through the shell fill the open positive-energy timelike cone. The quadratic polynomial  $q_1 - q_2$  therefore vanishes on a nonempty open subset of  $T_x^*M$ , and hence vanishes identically.  $\square$

This corollary is not an independent mechanism for producing the metric; it is the quadratic Lorentzian specialization of the familiar fact that a calibrated indicatrix determines its homogeneous norm. In the present construction the nontrivial step occurs earlier, when the internal shell is operationally soldered to  $T_x^*M$ .

The scalar response is invariant under changes of internal Lorentz frame. If

$$\Lambda_x \in SO^+(1, 3),$$

then

$$\Pi \mapsto \Lambda_x^{-T} \Pi, \quad e_x \mapsto \Lambda_x e_x, \quad (138)$$

and therefore

$$(\Lambda^{-T} \Pi)(\Lambda e \xi) = \Pi(e \xi).$$

Thus  $\Theta_x(\Pi, \xi)$  is independent of the chosen internal Lorentz frame, while

$$(\Lambda e)^* \eta = e^* \eta \quad (139)$$

leaves the reconstructed spacetime metric unchanged.

The absolute normalization of the response must be distinguished from its incidence structure. If operational access determines only the equivalence class

$$\Theta_x \sim \lambda(x) \Theta_x, \quad \lambda(x) > 0, \quad (140)$$

then

$$e_x \sim \lambda(x)e_x, \quad g_x \sim \lambda^2(x)g_x. \quad (141)$$

The null cone and conformal class are unchanged. An absolute generator–displacement calibration is required to transfer the fixed internal scale  $Q$  into an absolute spacetime metric scale. Thus the existence of the internal number  $Q$  alone does not remove the conformal ambiguity if the response itself is known only up to local positive rescaling.

No gravitational field equation has entered the argument. The result is kinematical and may be summarized as

$$\begin{array}{ccc} \text{internal relativistic massive-generator geometry} & & \\ + & & \\ \text{composition-compatible, calibrated, faithful} & \longrightarrow & e : TM \xrightarrow{\sim} V, \\ \text{incidence response} & & g = e^* \eta. \end{array} \quad (142)$$

The standard momentum–coframe pairing, the group-completion step, and the finite-dimensional representation of a bilinear pairing by a linear map are standard mathematical structures. What is specific to the present construction is their inverse operational organization: the scalar response is primitive only on the positive massive-generator sector, and the solder map is recovered from that response under explicit composition, regularity, calibration, and faithfulness conditions.

The next section addresses a separate dynamical question. Having reconstructed a Lorentzian soldered geometry, we ask which additional assumptions are sufficient to recover a conventional Einstein branch. That completion is conditional and does not follow from either the bounded MS relation or the soldering theorem alone.

## 5 Restricted Classical Gravitational Completion

Section 4 reconstructed a Lorentzian metric from the generator–displacement response. That result is kinematical: it determines the soldering of the internal Lorentzian generator geometry to spacetime, but it does not determine a gravitational field equation.

We now consider a restricted classical completion. The purpose is not to derive Einstein gravity from the MS state relation. Once Lorentzian soldering has been reconstructed, we enter standard first-order and canonical gravitational frameworks and ask what follows after a specified set of additional assumptions is imposed. The resulting Einstein branch is therefore conditional.

For the canonical analysis in this section we set

$$c = 1.$$

All action and Hamiltonian formulas below are to be read in these units unless explicitly stated otherwise.

### 5.1 Auxiliary torsion-free branch and metric phase space

Once the solder form has been reconstructed, the local geometry may be described by a coframe  $e^a$  and an independent metric-compatible Lorentz connection

$$\omega^{ab} = -\omega^{ba}. \quad (143)$$

Its torsion is

$$T^a := de^a + \omega^a_b \wedge e^b. \quad (144)$$

For a general matter action

$$S_m[e, \omega, \Psi],$$

define the Lorentz-spin current by the connection variation

$$\delta_\omega S_m = \frac{1}{2} \int_M \tau_{ab} \wedge \delta\omega^{ab}. \quad (145)$$

Here  $\tau_{ab}$  is the matter spin-current three-form, understood distributionally when the matter source is pointlike.

The structureless massive particle considered here has no spinor, Lorentz-bivector, or independent internal-orientation degree of freedom. In  $c = 1$  units its minimally coupled worldline action is

$$S_p = -m \int d\lambda \sqrt{-\eta_{ab} e^a_\mu e^b_\nu \dot{z}^\mu \dot{z}^\nu}. \quad (146)$$

The independent Lorentz connection does not occur in this action. Consequently,

$$\tau_{ab}^{\text{MS}} = 0. \quad (147)$$

Equation (147) is a statement about the matter source only. It does not imply  $T^a = 0$  in a general first-order, metric-affine, or gauge theory of gravity. Torsion may have propagating degrees of freedom or an unsourced homogeneous sector even in the absence of intrinsic matter spin [13, 11, 21]. We therefore impose an additional restriction.

**Assumption 5.1** (Auxiliary connection branch). *In the classical structureless sector considered here, the independent connection is auxiliary with respect to torsion: its field equation is local and algebraic in  $T^a$ , and for a nondegenerate coframe the unsourced equation has no nonzero torsional kernel. Schematically,*

$$\mathcal{M}_e(T) = \mathcal{J}(\tau), \quad (148)$$

with

$$\mathcal{M}_e(T) = 0 \implies T^a = 0. \quad (149)$$

Combining this assumption with Eq. (147) gives

$$T^a = 0. \quad (150)$$

Metric compatibility and vanishing torsion then determine the Levi–Civita spin connection uniquely,

$$\omega^{ab} = \omega^{ab}[e]. \quad (151)$$

Thus the torsion-free reduction is part of the definition of the restricted branch; it is not a consequence of the bounded MS state relation or of vanishing spin current alone.

Let

$$X : \Sigma \hookrightarrow M$$

be a spacelike hypersurface. Since  $c = 1$  in the present section, let  $n^a$  denote its future-directed internal unit normal,

$$\eta_{ab} n^a n^b = -1,$$

and choose an adapted Lorentz gauge

$$n^a = (1, 0, 0, 0).$$

Write the spatial pullback of the coframe as

$$E_i^I := e_i^I, \quad I = 1, 2, 3.$$

The induced spatial metric is

$$h_{ij} = \delta_{IJ} E_i^I E_j^J. \quad (152)$$

On the torsion-free branch the spatial pullback of the Lorentz connection decomposes as

$$\omega_i^I{}_J = \Gamma_i^I{}_J[E], \quad K_i^I := \omega_i^I{}_{\hat{0}}. \quad (153)$$

Defining

$$K_{ij} := K_i^I E_{jI}, \quad (154)$$

the torsion-free condition gives

$$K_{ij} = K_{ji}.$$

With the convention used here,

$$K_{ij} = h_i^\mu h_j^\nu \nabla_\mu n_\nu. \quad (155)$$

Thus  $K_{ij}$  is the standard extrinsic curvature. Its appearance in the boost components of the spin connection is the standard observer-adapted tetrad decomposition already anticipated in Section 3; no new definition of extrinsic curvature is introduced.

After reduction by the residual internal  $SO(3)$  gauge freedom, we restrict to the metric phase space

$$(h_{ij}, \pi^{ij}) \in T^* \text{Met}^+(\Sigma), \quad (156)$$

where  $\pi^{ij}$  is a symmetric contravariant tensor density of weight one. The canonical bracket is

$$\{h_{ij}(x), \pi^{kl}(y)\} = \delta_{(i}^k \delta_{j)}^l \delta^{(3)}(x - y). \quad (157)$$

No relation between  $\pi^{ij}$  and  $K_{ij}$  is assumed at this stage.

## 5.2 Restricted hypersurface-deformation completion

We now impose the standard Lorentzian hypersurface-deformation algebra as the canonical representation of normal and tangential hypersurface deformations [27, 17]. This algebra is not derived from the MS variables.

The spatial diffeomorphism generator is

$$\mathcal{D}[\mathbf{N}] = \int_{\Sigma} d^3x N^i \mathcal{H}_i, \quad (158)$$

with

$$\mathcal{H}_i = -2D_j \pi^j{}_i. \quad (159)$$

Here  $D_i$  is the Levi-Civita derivative of  $h_{ij}$ , acting on tensor densities with the appropriate density weight.

For

$$\mathcal{H}[N] = \int_{\Sigma} d^3x N \mathcal{H}_{\perp},$$

the target algebra is

$$\{\mathcal{D}[\mathbf{N}], \mathcal{D}[\mathbf{M}]\} = \mathcal{D}[[\mathbf{N}, \mathbf{M}]], \quad (160)$$

$$\{\mathcal{D}[\mathbf{N}], \mathcal{H}[M]\} = \mathcal{H}[\mathcal{L}_{\mathbf{N}}M], \quad (161)$$

$$\{\mathcal{H}[N], \mathcal{H}[M]\} = \mathcal{D}[\xi], \quad (162)$$

where

$$\xi^i = h^{ij} (N \partial_j M - M \partial_j N). \quad (163)$$

The role of the MS construction is upstream. The operational soldering theorem has supplied a Lorentzian tangent geometry to which the standard hypersurface description can be applied. Refoliation consistency can then be imposed in its usual canonical form. Nothing in Eqs. (160)–(162) is claimed as an independently derived MS algebra.

To obtain a definite metric dynamics, we restrict the scalar constraint to the separable form

$$\mathcal{H}_{\perp} = \mathcal{T}(h, \pi) + \mathcal{V}(h, \partial h, \partial^2 h), \quad (164)$$

with the following assumptions:

1.  $\mathcal{H}_{\perp}$  is a local spatially diffeomorphism-covariant scalar density of weight one;
2.  $\mathcal{T}$  is even and quadratic in  $\pi^{ij}$ , contains no spatial derivatives of  $\pi^{ij}$ , and its kinetic supermetric is constructed ultralocally from  $h_{ij}$  alone;
3. the kinetic and potential sectors are separated: mixed curvature–momentum couplings such as

$${}^{(3)}R \pi^{ij} \pi_{ij}, \quad {}^{(3)}R_{ij} \pi^{ik} \pi^j_k$$

are excluded;

4.  $\mathcal{V}$  is independent of  $\pi^{ij}$ , contains at most two spatial derivatives of  $h_{ij}$ , and is at most linear in the spatial curvature, modulo a spatial boundary term.

These are defining restrictions of the canonical branch studied here. In particular, they do not follow merely from locality, time-reversal symmetry, spatial covariance, or the MS state geometry.

Within this class, the scalar constraint has the form

$$\mathcal{H}_{\perp} = \frac{A}{\sqrt{h}} (\pi^{ij} \pi_{ij} - \alpha \pi^2) - B \sqrt{h} {}^{(3)}R + C \sqrt{h}, \quad (165)$$

where

$$\pi := h_{ij} \pi^{ij},$$

and  $A, B, C, \alpha$  are constants in the restricted branch.

The explicit calculation in Appendix B gives

$$\begin{aligned} \{\mathcal{H}[N], \mathcal{H}[M]\} &= AB \mathcal{D}[\xi] \\ &\quad - 2AB(2\alpha - 1) \int_{\Sigma} d^3x \xi^i D_i \pi. \end{aligned} \quad (166)$$

The second term is not proportional to the momentum constraint on the unrestricted metric phase space.

The target bracket Eq. (162) contains a nonvanishing diffeomorphism term for generic phase-space data and smearing functions. Consequently, its realization requires

$$AB \neq 0. \quad (167)$$

The independent trace-gradient obstruction in Eq. (166) must then vanish off shell, without imposing an additional phase-space restriction such as  $D_i \pi = 0$ . Hence

$$\alpha = \frac{1}{2}. \quad (168)$$

The remaining bracket becomes

$$\{\mathcal{H}[N], \mathcal{H}[M]\} = AB \mathcal{D}[\xi],$$

and matching the normalization of Eq. (162) gives

$$AB = 1. \quad (169)$$

The ultralocal coefficient  $C$  is not constrained by this derivative-of-smearing calculation.

Introduce a nonzero constant  $\kappa$  and a constant  $\Lambda$  by

$$A = 2\kappa, \quad B = \frac{1}{2\kappa}, \quad C = \frac{\Lambda}{\kappa}. \quad (170)$$

The conventional branch has  $\kappa > 0$ . The scalar constraint is then

$$\mathcal{H}_\perp = \frac{2\kappa}{\sqrt{h}} \left( \pi^{ij} \pi_{ij} - \frac{1}{2} \pi^2 \right) - \frac{\sqrt{h}}{2\kappa} \left( {}^{(3)}R - 2\Lambda \right), \quad (171)$$

together with Eq. (159). These are the ADM constraints in the conventions adopted here.

Hamilton's equation for  $h_{ij}$  is

$$\dot{h}_{ij} = \frac{4\kappa N}{\sqrt{h}} \left( \pi_{ij} - \frac{1}{2} \pi h_{ij} \right) + D_i \beta_j + D_j \beta_i. \quad (172)$$

With the extrinsic-curvature convention Eq. (155), the kinematical decomposition is

$$\dot{h}_{ij} = 2NK_{ij} + D_i \beta_j + D_j \beta_i. \quad (173)$$

Comparison gives

$$\pi^{ij} = \frac{\sqrt{h}}{2\kappa} \left( K^{ij} - K h^{ij} \right). \quad (174)$$

The canonical Legendre transform yields

$$S_{\text{ADM}} = \frac{1}{2\kappa} \int dt d^3x N \sqrt{h} \left[ {}^{(3)}R + K_{ij} K^{ij} - K^2 - 2\Lambda \right]. \quad (175)$$

The Gauss–Codazzi identity then gives, up to the standard boundary contribution,

$$S_{\text{EH}} = \frac{1}{2\kappa} \int_M d^4x \sqrt{-g} \left( {}^{(4)}R - 2\Lambda \right). \quad (176)$$

Equations (175) and (176) remain written in the  $c = 1$  convention adopted at the beginning of this section.

With minimally coupled matter, the corresponding field equation is

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu} \quad (c = 1). \quad (177)$$

When conventional physical units are restored, the coupling appearing in the Einstein field equation is

$$\kappa_{\text{phys}} = \frac{8\pi G}{c^4}. \quad (178)$$

We do not restore powers of  $c$  term by term in the preceding canonical action formulas. In particular, Eq. (176) should not be combined directly with Eq. (178) without simultaneously restoring the time-coordinate and action conventions.

Neither the numerical value of  $G$  nor the cosmological constant  $\Lambda$  is determined by the MS state geometry, the operational soldering theorem, or the canonical closure calculation.

The logical status of this completion can now be stated compactly.

**Proposition 5.2** (Restricted minimal Einstein completion). *Assume that:*

1. *the hypotheses of Theorem 4.6 hold, so that a Lorentzian soldered metric  $g = e^*\eta$  has been reconstructed;*
2. *the structureless matter sector has vanishing intrinsic Lorentz-spin current and the independent connection lies in the auxiliary branch of Assumption 5.1;*
3. *after torsion and local Lorentz gauge reduction, the canonical phase space is  $(h_{ij}, \pi^{ij})$ ;*
4. *the canonical generators realize the standard Lorentzian hypersurface-deformation algebra off shell; and*
5. *the scalar constraint belongs to the restricted separable class defined by Eq. (164) and the four restrictions stated above.*

*Then*

$$\alpha = \frac{1}{2}, \quad AB = 1, \quad (179)$$

*and, after Eq. (170), the resulting metric canonical theory is the ADM theory with free constants  $\kappa$  and  $\Lambda$ . The standard Legendre transform and Gauss–Codazzi identity yield the Einstein–Hilbert bulk action up to the appropriate boundary term.*

This proposition has a different logical status from Theorem 4.6. The latter concerns the operational reconstruction of the internal–tangent incidence map. The present proposition begins only after a metric phase space, an auxiliary torsion-free branch, the standard hypersurface-deformation algebra, a derivative order, and a kinetic–potential separation have been imposed. It is therefore a restricted HKT/ADM completion rather than a general derivation or uniqueness theorem for relativistic gravity.

Relaxing these assumptions can lead outside the Einstein branch. Intrinsic spin, independent or propagating torsion, curvature–momentum couplings, higher-curvature terms, higher powers or spatial derivatives of the canonical momentum, additional fields, nonlocal generators, enlarged phase spaces, or alternative generally covariant canonical realizations are not excluded by the MS geometry or by the soldering theorem.

## 6 Discussion and Conclusions

The construction developed in this paper contains two logically distinct layers. The first is an inverse kinematical problem: an internal Lorentzian massive-generator geometry is assumed independently of the tangent bundle, and a scalar generator–displacement response is used to reconstruct their soldering. The second is a restricted gravitational completion performed only after that Lorentzian geometry has been obtained. These layers should be kept separate: neither the bounded MS state relation nor the operational soldering theorem by itself determines a gravitational field equation.

### 6.1 Logical status, relation to existing structures, and principal result

The local MS state

$$(\mathbf{D}, S), \quad |\mathbf{D}|^2 + S^2 = 1, \quad S > 0, \quad (180)$$

is a bounded representation of the standard future-timelike state space. The physical upper hemisphere is not compact; its closure is a compactification obtained by adjoining the infinite-rapidity boundary. The projective map

$$\Phi(\mathbf{D}, S) = (S^{-1}, S^{-1}\mathbf{D}) \quad (181)$$

identifies the physical  $S > 0$  sector diffeomorphically with the future unit timelike hyperboloid. Consequently, once a Lorentzian spacetime metric is given, the observer bundle, Lorentz connection, curvature, tetrad decomposition, and boost–rotation splitting appearing in Sections 2 and 3 are standard Lorentzian geometric structures. The MS variables provide a nonlinear bounded coordinate representation of these structures; they do not define an independent local causal geometry.

The companion flat-space analysis [9] supplies two logically prior ingredients. Its operational realization identifies the Minkowski timelike structure of flat translation space, while its rigidity results select the standard relativistic free-particle dispersion relation, massive generator shell, and connected Lorentz observer structure. The present paper imports this relativistic generator geometry but removes the global affine translation structure on which the flat construction relies. On a generic curved manifold there is no canonical global Abelian translation space. Instead, we begin with

$$TM \quad \text{and} \quad V \quad (182)$$

as independent bundles and ask how the internal generator geometry acquires tangent-space meaning.

The generator relevant to this inverse problem is the Lorentz covector

$$\Pi \in V_x^*,$$

rather than the contravariant four-momentum vector  $P \in V_x$ . The primitive curved-space datum is

$$\Theta_x(\Pi, \xi), \quad \Pi \in C_{Q,x}, \quad \xi \in T_x M. \quad (183)$$

Generator additivity and positive homogeneity, displacement additivity and continuity, and smooth base-point dependence extend Eq. (183) to a bilinear incidence pairing

$$\widetilde{\Theta}_x : V_x^* \times T_x M \longrightarrow \mathbb{R}. \quad (184)$$

Operational faithfulness states that no nonzero tangent displacement is invisible to the physical massive-generator sector. In the equal-rank four-dimensional setting this is equivalent to nondegeneracy of the resulting map

$$e : TM \xrightarrow{\sim} V, \quad \widetilde{\Theta}_x(\Pi, \xi) = \Pi(e_x \xi). \quad (185)$$

The Lorentzian metric is then

$$g = e^* \eta. \quad (186)$$

Several ingredients in this chain are standard. Once  $e$  is known, the pairing

$$\Pi(e\xi)$$

is the usual momentum–coframe incidence familiar from tetrad and first-order formulations. Extending an additive positively homogeneous map from a generating cone is a standard group-completion argument, and representing the resulting bilinear form by  $e_x \in \text{Hom}(T_x M, V_x)$  is elementary finite-dimensional duality. Likewise, the relation between causal structure and conformal Lorentzian geometry is classical [4, 10, 20], and the determination of a homogeneous geometry by a calibrated mass shell or indicatrix has close analogues in Finsler and Hamiltonian dispersion geometry [23, 2].

The specific reconstruction result lies in the choice of primitive data. The vector-valued solder map is not supplied at the outset. Instead, scalar responses are specified only for physically admissible positive massive generators and infinitesimal tangent displacements. The composition and regularity assumptions recover the full bilinear incidence structure, while operational faithfulness makes the resulting solder map nondegenerate. The result is therefore an *inverse operational characterization of Lorentzian soldering*.

This formulation also distinguishes the present construction from other reconstruction programs. In the Ehlers–Pirani–Schild approach, light rays and freely falling trajectories constrain conformal and projective spacetime structures [4]. Premetric electrodynamics reconstructs light-cone and metric information from electromagnetic constitutive structure [12, 24]. General dispersion-relation programs infer admissible spacetime geometries from the propagation properties of matter [23, 16]. Observer-space and Cartan-geometric approaches may instead treat observer space as fundamental and, under suitable integrability conditions, reconstruct spacetime as a quotient [6, 14]. Here the manifold  $M$  is already given, as is an independent internal Lorentz bundle  $V$ ; the reconstructed object is the nondegenerate identification

$$TM \simeq V$$

obtained from a massive generator–displacement incidence response.

Related Cartan-geometric, gauge-gravity, and first-order formulations provide established ways of organizing coframe, connection, and metric variables [26, 6, 13]. The distinction of the present construction is the operational origin assigned to the soldering data, rather than the general possibility of using nonmetric or noncoframe primitive variables.

The cone and the fixed massive shell play different roles. The positive timelike cone

$$C_{Q,x}$$

is independent of the numerical value of  $Q$  and, after soldering, determines the positive timelike cotangent cone and hence the pointwise causal and conformal structure. The calibrated fixed shell

$$\eta^*(\Pi, \Pi) = -Q^2 \quad (187)$$

is transferred to

$$g^{-1}(p, p) = -Q^2. \quad (188)$$

It carries the absolute quadratic normalization. If instead

$$\Theta_x \sim \lambda(x)\Theta_x, \quad \lambda(x) > 0,$$

then

$$e_x \sim \lambda(x)e_x, \quad g_x \sim \lambda^2(x)g_x,$$

and only the conformal class is reconstructed. Thus an internal scale  $Q$  fixes the spacetime metric scale only when the generator–displacement response is itself absolutely calibrated.

The global statement is likewise conditional. If a globally smooth faithful response exists, the resulting solder map implies

$$TM \simeq V. \quad (189)$$

The existence of such a global bundle isomorphism is a topological compatibility condition encoded in the hypotheses. The local theorem does not assert that an arbitrary four-manifold and an arbitrary Lorentz vector bundle necessarily satisfy this condition.

The gravitational completion has a different logical status. The standard hypersurface-deformation algebra and the Hojman–Kuchař–Teitelboim inverse problem are established canonical structures [27, 17]. Related programs have also studied how matter dynamics and causal structure constrain compatible gravitational dynamics [7, 3]. Section 5 does not attempt a general classification. It assumes an auxiliary torsion-free branch, reduces to the metric phase space, imposes the standard Lorentzian hypersurface-deformation algebra, and restricts the scalar constraint to a separable low-derivative quadratic form without curvature–momentum couplings. Within that class, off-shell closure gives

$$\alpha = \frac{1}{2}, \quad AB = 1, \quad (190)$$

and hence the ADM form with free  $\kappa$  and  $\Lambda$ . The logical implication is

$$\begin{array}{ccc} \text{reconstructed Lorentzian soldering} & & \\ + & \longrightarrow & \text{ADM/Einstein branch.} \\ \text{restricted canonical assumptions} & & \end{array} \quad (191)$$

This is a restricted completion rather than a general uniqueness theorem for relativistic gravitational dynamics.

## 6.2 Scope, limitations, and outlook

The assumptions delimit the scope of the reconstruction.

First, neither spacetime dimension nor Lorentzian signature is derived in this paper. We assume

$$\dim M = 4, \quad \text{rank } V = 4, \quad (192)$$

and the internal bundle is already equipped with the Lorentzian form selected by the preceding flat-space theory. The present inverse problem concerns soldering that internal geometry to tangent space, not the independent derivation of dimension or signature.

Second, the classical response used here is built from the massive timelike generator sector. At fixed  $Q > 0$ ,

$$S \rightarrow 0^+$$

is an infinite-rapidity and infinite-energy boundary of a massive shell; it is not an ordinary massive state becoming null. The companion flat-space theory admits a distinct scale-free null completion through the finite-energy  $Q \rightarrow 0^+$  limit or a homogeneous lift of the projective boundary. Extending

the present generator–displacement soldering construction to that null sector, and further to general massless fields, requires additional structure.

Third, the matter sector considered here is structureless and spinless. Its worldline action contains no independent Lorentz connection and hence no intrinsic Lorentz-spin current. This does not imply vanishing torsion in a general gravitational theory. The torsion-free branch of Section 5 follows only after the independent auxiliary-connection assumption. Spinorial matter, propagating torsion, metric-affine degrees of freedom, and more general gauge-gravity sectors remain outside the present construction.

Fourth, the restricted Einstein completion is not stable under arbitrary enlargement of its canonical ansatz. Curvature–momentum couplings, higher-curvature potentials, higher powers or spatial derivatives of  $\pi^{ij}$ , additional canonical fields, nonlocal generators, and alternative phase spaces are not excluded by the MS state geometry or by the soldering theorem. Section 5 therefore provides a minimal conventional completion within a specified canonical class, rather than a classification of generally covariant dynamics.

Fifth, bounded MS coordinates do not introduce a physical ultraviolet cutoff. For a massive species,

$$\mathbf{p} = mc \frac{\mathbf{D}}{S}, \quad (193)$$

so the physical momentum remains unbounded as  $S \rightarrow 0^+$ . The bounded representation compactifies an infinite-rapidity direction in state coordinates; it does not imply a maximal momentum, a minimal length, or ultraviolet regularization.

Sixth, two gravitational parameters remain undetermined. Neither the operational soldering theorem nor the restricted canonical closure fixes the gravitational coupling. In conventional units the Einstein equation contains

$$\kappa_{\text{phys}} = \frac{8\pi G}{c^4}, \quad (194)$$

while the cosmological constant  $\Lambda$  also remains free. Their values require input beyond the classical state–response–soldering structure considered here.

Finally, the present construction is classical. Associating a phase with a response,

$$\exp \left[ -\frac{i}{\hbar} \Theta_x(\Pi, \xi) \right], \quad (195)$$

does not by itself produce a noncommutative quantum geometry. In particular, exact additive implementation of the same scalar response need not generate noncommuting local observables. A quantum extension requires additional configuration, conjugate, comparison, or projective structure rather than the formal insertion of  $\hbar$ .

Within these qualifications, the principal result is the classical inverse reconstruction

$$\begin{array}{ccc} \text{rigidity-selected massive Lorentz generator geometry} & & \\ + & & \\ \text{calibrated, composition-compatible, faithful} & \longrightarrow & e : TM \xrightarrow{\sim} V, \\ \text{generator–displacement response} & & g = e^* \eta. \end{array} \quad (196)$$

The information contained in the extended response is ultimately equivalent to that of a solder map. The purpose of the construction is therefore not to reduce the number of local geometric degrees

of freedom, but to specify those degrees of freedom operationally through scalar massive-generator incidences before a coframe is introduced.

Several extensions are natural. For isolated systems, noncompact hypersurfaces lead to the Hamiltonian boundary problem and asymptotic translation charges. Stationary and axisymmetric geometries permit the response to be restricted to symmetry-generated tangent directions, providing a natural setting for redshift, frame dragging, ergosurfaces, Killing horizons, and stationary charges. A quantum extension would instead require promoting the reconstructed soldering data to suitable configuration and conjugate variables. Each direction requires additional asymptotic, symmetry, or quantum assumptions beyond those used in the present paper.

The framework established here is therefore an operational reconstruction of the internal–tangent soldering relation from the positive massive-generator sector. Under the separately stated canonical assumptions, it also admits a conventional restricted classical gravitational completion. The central inverse result, however, is the reconstruction

$$\Theta \longrightarrow e \longrightarrow g = e^* \eta,$$

which identifies the operational conditions under which an independent internal relativistic generator geometry acquires spacetime tangent-space meaning.

## A Technical Foundations of the Operational Soldering Theorem

This appendix collects the algebraic and differential details underlying Theorem 4.6. The points requiring explicit verification are the extension of the response from the positive massive generator cone to the full internal dual space, the equivalence between operational faithfulness and solder-map nondegeneracy, and the smoothness and normalization properties of the reconstructed bundle map.

Throughout,  $V_x$  is a four-dimensional oriented and time-oriented Lorentz vector space with metric signature

$$(-, +, +, +),$$

and  $V_x^*$  is its dual. For  $Q > 0$ , define the positive-energy massive covector shell

$$\Sigma_{Q,x}^+ = \left\{ \Pi \in V_x^* : \eta_x^*(\Pi, \Pi) = -Q^2, \Pi \text{ on the positive-energy sheet} \right\}. \quad (197)$$

In an oriented and time-oriented orthonormal dual frame  $\{\varepsilon^0, \varepsilon^1, \varepsilon^2, \varepsilon^3\}$ , our convention is

$$E = -c \Pi_0 > 0.$$

Thus the rest covector is

$$-Q \varepsilon^0.$$

### A.1 The positive generator cone and linear extension

We first show that one fixed positive massive shell already generates the entire internal dual space.

**Lemma A.1** (Positive-shell basis). *The shell  $\Sigma_{Q,x}^+$  contains four linearly independent covectors. Consequently,*

$$\text{span } \Sigma_{Q,x}^+ = V_x^*. \quad (198)$$

*Proof.* Fix  $a > 0$  and set  $E_a := \sqrt{Q^2 + a^2}$ . The four covectors

$$\Pi^{(0)} = -Q \varepsilon^0, \quad (199)$$

$$\Pi^{(1)} = -E_a \varepsilon^0 + a \varepsilon^1, \quad (200)$$

$$\Pi^{(2)} = -E_a \varepsilon^0 + a \varepsilon^2, \quad (201)$$

$$\Pi^{(3)} = -E_a \varepsilon^0 + a \varepsilon^3 \quad (202)$$

all belong to  $\Sigma_{Q,x}^+$ .

If

$$\sum_{A=0}^3 c_A \Pi^{(A)} = 0,$$

the three spatial components imply

$$c_1 = c_2 = c_3 = 0,$$

and the temporal component then gives  $c_0 = 0$ . Hence the four covectors are linearly independent.  $\square$

Define the positive generator cone

$$C_{Q,x} := \left\{ \sum_{A=1}^N \lambda_A \Pi_A : N < \infty, \lambda_A \geq 0, \Pi_A \in \Sigma_{Q,x}^+ \right\}. \quad (203)$$

**Lemma A.2** (Mass independence of the positive cone). *For every  $Q > 0$ ,*

$$C_{Q,x} = \{0\} \cup \{\Pi \in V_x^* : \eta_x^*(\Pi, \Pi) < 0, \Pi \text{ on the positive-energy side}\}. \quad (204)$$

*In particular, the cone is independent of the numerical value of  $Q$ .*

*Proof.* Let  $\Pi \neq 0$  be any positive-energy timelike covector and define

$$r(\Pi) := \sqrt{-\eta_x^*(\Pi, \Pi)}. \quad (205)$$

Then

$$\widehat{\Pi} := \frac{Q}{r(\Pi)} \Pi \in \Sigma_{Q,x}^+, \quad (206)$$

and

$$\Pi = \frac{r(\Pi)}{Q} \widehat{\Pi} \in C_{Q,x}. \quad (207)$$

Conversely, a nonzero positive linear combination of positive-energy timelike covectors is again positive-energy timelike, because the future timelike cone of a time-oriented Lorentz vector space is convex. This proves Eq. (204).  $\square$

Since the cone is generating,

$$C_{Q,x} - C_{Q,x} = V_x^*. \quad (208)$$

Equivalently, this follows from Lemma A.1. The role of  $C_{Q,x}$  is algebraic: it supplies the additive positive domain on which the primitive response is defined. Elements away from  $\Sigma_{Q,x}^+$  are not thereby interpreted as one-particle states of fixed rest mass  $Q/c$ .

Let

$$\Theta_x : C_{Q,x} \times T_x M \longrightarrow \mathbb{R} \quad (209)$$

satisfy the composition assumptions of Section 4. Thus, for fixed  $\xi$ ,

$$\Theta_x(\Pi_1 + \Pi_2, \xi) = \Theta_x(\Pi_1, \xi) + \Theta_x(\Pi_2, \xi), \quad (210)$$

$$\Theta_x(\lambda \Pi, \xi) = \lambda \Theta_x(\Pi, \xi), \quad \lambda \geq 0, \quad (211)$$

and, for fixed  $\Pi$ ,

$$\Theta_x(\Pi, \xi + \zeta) = \Theta_x(\Pi, \xi) + \Theta_x(\Pi, \zeta), \quad (212)$$

with continuity in the displacement argument.

For

$$\Pi = \Pi_+ - \Pi_-, \quad \Pi_{\pm} \in C_{Q,x},$$

define

$$\widetilde{\Theta}_x(\Pi, \xi) := \Theta_x(\Pi_+, \xi) - \Theta_x(\Pi_-, \xi). \quad (213)$$

**Lemma A.3** (Well-defined generator extension). *Equation (213) is independent of the decomposition  $\Pi = \Pi_+ - \Pi_-$  and defines the unique real-linear extension of the generator response from  $C_{Q,x}$  to  $V_x^*$ .*

*Proof.* Suppose  $\Pi = \Pi_+ - \Pi_- = \Pi'_+ - \Pi'_-$ . Then  $\Pi_+ + \Pi'_- = \Pi'_+ + \Pi_-$ . Additivity on the cone gives

$$\begin{aligned} \Theta_x(\Pi_+, \xi) + \Theta_x(\Pi'_-, \xi) &= \Theta_x(\Pi_+ + \Pi'_-, \xi) \\ &= \Theta_x(\Pi'_+ + \Pi_-, \xi) \\ &= \Theta_x(\Pi'_+, \xi) + \Theta_x(\Pi_-, \xi), \end{aligned} \quad (214)$$

which proves independence of the chosen decomposition.

Additivity extends immediately to differences of cone elements. For  $\lambda > 0$ , positive homogeneity gives

$$\widetilde{\Theta}_x(\lambda \Pi, \xi) = \lambda \widetilde{\Theta}_x(\Pi, \xi),$$

while

$$\widetilde{\Theta}_x(-\Pi, \xi) = -\widetilde{\Theta}_x(\Pi, \xi)$$

follows by interchanging  $\Pi_+$  and  $\Pi_-$ . Hence the extension is real linear in  $\Pi$ . Uniqueness follows because  $C_{Q,x}$  generates  $V_x^*$ .  $\square$

For fixed  $\Pi \in C_{Q,x}$ , displacement additivity and continuity imply the continuous Cauchy relation

$$\Theta_x(\Pi, \lambda \xi) = \lambda \Theta_x(\Pi, \xi), \quad \lambda \in \mathbb{R}. \quad (215)$$

The same relation then holds for the extension  $\widetilde{\Theta}_x$ . Consequently,

$$\widetilde{\Theta}_x : V_x^* \times T_x M \longrightarrow \mathbb{R} \quad (216)$$

is bilinear.

This is the point at which the primitive response on the positive generator sector becomes an ordinary bilinear incidence pairing. The extension is the standard group completion of an additive response on a generating cone.

## A.2 Faithfulness, smoothness, and normalization

For every  $\xi \in T_x M$ , bilinearity makes

$$\Pi \mapsto \widetilde{\Theta}_x(\Pi, \xi)$$

an element of  $V_x^{**}$ . Using the canonical finite-dimensional identification  $V_x^{**} \simeq V_x$ , define  $e_x \xi \in V_x$  uniquely by

$$\Pi(e_x \xi) = \widetilde{\Theta}_x(\Pi, \xi) \quad \forall \Pi \in V_x^*. \quad (217)$$

Linearity of  $\widetilde{\Theta}_x$  in  $\xi$  makes

$$e_x : T_x M \longrightarrow V_x \quad (218)$$

a linear map.

Define the operationally invisible tangent subspace

$$\mathcal{N}_x := \left\{ \xi \in T_x M : \Theta_x(\Pi, \xi) = 0 \quad \forall \Pi \in \Sigma_{Q,x}^+ \right\}. \quad (219)$$

**Proposition A.4** (Faithfulness and solder-map nondegeneracy). *Under the composition assumptions,*

$$\mathcal{N}_x = \ker e_x. \quad (220)$$

*Hence operational faithfulness is equivalent to injectivity of  $e_x$ . Since  $\dim T_x M = \dim V_x = 4$ , it is equivalent in the present setting to  $e_x$  being a vector-space isomorphism.*

*Proof.* If  $\xi \in \mathcal{N}_x$ , the response vanishes on the shell. Since the shell spans  $V_x^*$  and the extension is linear,  $\widetilde{\Theta}_x(\Pi, \xi) = 0$  for every  $\Pi \in V_x^*$ . Equation (217) then implies  $\Pi(e_x \xi) = 0$  for every  $\Pi \in V_x^*$ , and therefore  $e_x \xi = 0$ . Thus  $\mathcal{N}_x \subseteq \ker e_x$ .

Conversely, if  $e_x \xi = 0$ , then

$$\Theta_x(\Pi, \xi) = \Pi(e_x \xi) = 0$$

for every shell covector  $\Pi$ . Hence  $\xi \in \mathcal{N}_x$ , so  $\ker e_x \subseteq \mathcal{N}_x$ .  $\square$

This proposition makes the role of operational faithfulness explicit. Once the response has been extended bilinearly, distinguishability of nonzero tangent displacements and nondegeneracy of the solder map are equivalent. The distinction lies in the primitive formulation: the former is imposed on scalar generator responses, whereas the latter would be imposed directly on a pre-existing coframe.

We next verify smoothness. Assume that for every smooth local shell or cone section

$$\Pi : U \longrightarrow C_Q$$

and every smooth vector field

$$\xi \in \Gamma(TU),$$

the scalar function

$$x \mapsto \Theta_x(\Pi(x), \xi(x)) \quad (221)$$

is smooth.

Fix  $x_0 \in M$ . On a sufficiently small neighborhood  $U$ , choose four smooth shell sections

$$\Pi^{(A)}(x), \quad A = 0, 1, 2, 3, \quad (222)$$

that remain linearly independent throughout  $U$ . Such sections exist by choosing a smooth local oriented and time-oriented Lorentz frame and using the construction of Lemma A.1. Let  $E_A(x)$  be the local frame of  $V$  dual to  $\Pi^{(A)}(x)$ . Then

$$e_x \xi = \sum_{A=0}^3 \Theta_x(\Pi^{(A)}, \xi) E_A(x). \quad (223)$$

Each coefficient is smooth by hypothesis. Therefore

$$e \in \Gamma(T^*M \otimes V) \quad (224)$$

is smooth.

The construction is intrinsically local. If the response is defined globally and obeys the same smoothness and faithfulness conditions, the locally reconstructed maps agree on overlaps by uniqueness and define a global bundle isomorphism

$$e : TM \xrightarrow{\sim} V. \quad (225)$$

Consequently,

$$TM \simeq V \quad (226)$$

is a necessary topological condition for the existence of the globally faithful response assumed in the global form of Theorem 4.6. The theorem does not assert that an arbitrary pair  $(TM, V)$  admits such a response.

The reconstruction may also be displayed using four independent massive probes. Let

$$\Pi^{(A)} \in \Sigma_{Q,x}^+, \quad A = 0, 1, 2, 3, \quad (227)$$

form a basis of  $V_x^*$ , and define

$$\theta^{(A)}(\xi) := \Theta_x(\Pi^{(A)}, \xi). \quad (228)$$

Then

$$\theta^{(A)} = e_x^* \Pi^{(A)}. \quad (229)$$

If the response is faithful, the four  $\theta^{(A)}$  are linearly independent and hence form a coframe of  $T_x^*M$ . If  $E_A$  is the basis of  $V_x$  dual to  $\Pi^{(A)}$ , then

$$e_x = \sum_{A=0}^3 E_A \otimes \theta^{(A)}. \quad (230)$$

This finite-probe formula represents the reconstructed map; it does not replace either the composition assumptions or operational faithfulness.

Finally, consider normalization. If the primitive response is known only up to the local equivalence relation

$$\Theta_x \sim \lambda(x) \Theta_x, \quad \lambda(x) > 0, \quad (231)$$

then the corresponding extensions satisfy

$$\widetilde{\Theta}_x \mapsto \lambda(x) \widetilde{\Theta}_x,$$

and therefore

$$e_x \mapsto e'_x = \lambda(x) e_x. \quad (232)$$

The reconstructed metric consequently transforms as

$$g_x \mapsto g'_x = \lambda^2(x) g_x. \quad (233)$$

**Proposition A.5** (Conformal reconstruction). *If the generator–displacement response is operationally determined only up to a smooth positive local normalization, then the reconstruction determines only the conformal class*

$$[g] = \{\Omega^2 g : \Omega > 0\}. \quad (234)$$

*Proof.* Equation (233) shows that different representatives of the same response-normalization class produce metrics related by a smooth positive conformal rescaling. Conversely, once a representative of the response is absolutely calibrated, Theorem 4.6 fixes  $e$ , and therefore  $g = e^* \eta$ , uniquely.  $\square$

The positive timelike cone and its null boundary are unchanged by the rescaling  $g \mapsto \lambda^2 g$ . Absolute proper-time normalization and the numerical normalization of the physical fixed- $Q$  shell are not. Thus the internal shell scale  $Q$  acquires an absolute spacetime interpretation only after the generator–displacement response itself has been calibrated.

## B Restricted Canonical Closure

This appendix gives the functional calculation underlying Eq. (166). Its scope is restricted to the canonical class specified in Section 5. The kinetic term is ultralocal in the spatial metric, quadratic in the canonical momentum, and contains no spatial derivatives of that momentum; the potential is independent of the momentum and is restricted to the scalar-curvature term plus an ultralocal term. Mixed curvature–momentum couplings are excluded by assumption.

We assume either that  $\Sigma$  is compact without boundary or that all smearing functions have compact support. Boundary terms produced by the integrations by parts below may therefore be discarded.

The canonical variables are

$$(h_{ij}, \pi^{ij}), \quad (235)$$

where  $h_{ij}$  is a positive-definite spatial metric and  $\pi^{ij} = \pi^{ji}$  is a contravariant tensor density of weight one. Their Poisson bracket is

$$\{h_{ij}(x), \pi^{kl}(y)\} = \delta_i^k \delta_j^l \delta^{(3)}(x - y). \quad (236)$$

The Levi–Civita derivative  $D_i$  associated with  $h_{ij}$  is understood to act on  $\pi^{ij}$  and its contractions with the appropriate density weight.

The spatial diffeomorphism generator is

$$\mathcal{D}[\mathbf{N}] = \int_{\Sigma} d^3x N^i \mathcal{H}_i, \quad (237)$$

with

$$\mathcal{H}_i = -2D_j \pi^j_i. \quad (238)$$

The scalar generator is

$$\mathcal{H}[N] = \int_{\Sigma} d^3x N \mathcal{H}_{\perp}, \quad (239)$$

where

$$\mathcal{H}_{\perp} = \frac{A}{\sqrt{h}} (\pi^{ij} \pi_{ij} - \alpha \pi^2) - B \sqrt{h} {}^{(3)}R + C \sqrt{h}, \quad (240)$$

and

$$\pi := h_{ij} \pi^{ij}.$$

Within the branch considered here,  $A, B, C, \alpha$  are constants.

The target normal–normal Lorentzian hypersurface-deformation relation is

$$\{\mathcal{H}[N], \mathcal{H}[M]\} = \mathcal{D}[\xi], \quad (241)$$

where

$$\xi^i = h^{ij} (N \partial_j M - M \partial_j N). \quad (242)$$

## B.1 Functional calculation and the trace obstruction

Decompose

$$\mathcal{H}[N] = K[N] + V_R[N] + V_0[N], \quad (243)$$

with

$$K[N] = A \int_{\Sigma} d^3x \frac{N}{\sqrt{h}} (\pi^{ij} \pi_{ij} - \alpha \pi^2), \quad (244)$$

$$V_R[N] = -B \int_{\Sigma} d^3x N \sqrt{h} {}^{(3)}R, \quad (245)$$

$$V_0[N] = C \int_{\Sigma} d^3x N \sqrt{h}. \quad (246)$$

Define

$$P_{ij} := \pi_{ij} - \alpha \pi h_{ij}. \quad (247)$$

The derivative of the kinetic term with respect to the momentum is

$$\frac{\delta K[N]}{\delta \pi^{ij}} = \frac{2AN}{\sqrt{h}} P_{ij}. \quad (248)$$

For a nonconstant smearing function  $M$ , variation of the curvature functional with respect to the covariant spatial metric gives

$$\begin{aligned} \delta \int_{\Sigma} d^3x M \sqrt{h} {}^{(3)}R &= \int_{\Sigma} d^3x \sqrt{h} \left[ -M {}^{(3)}G^{ij} \right. \\ &\quad \left. + D^i D^j M - h^{ij} D^2 M \right] \delta h_{ij}, \end{aligned} \quad (249)$$

where

$$D^2 := D_i D^i.$$

Only terms containing derivatives of the smearing function can survive antisymmetrization under  $N \leftrightarrow M$ . Hence

$$\left. \frac{\delta V_R[M]}{\delta h_{ij}} \right|_{\partial M} = -B \sqrt{h} (D^i D^j M - h^{ij} D^2 M). \quad (250)$$

The metric derivative of  $K[N]$  is ultralocal in  $N$ , while  $V_0[N]$  contains no derivatives of its smearing function. Their antisymmetrized contributions therefore cancel. The kinetic–kinetic bracket vanishes identically because  $K$  contains no spatial derivatives, and the potential–potential bracket vanishes because both potentials are independent of  $\pi^{ij}$ . Thus

$$\{\mathcal{H}[N], \mathcal{H}[M]\} = \{K[N], V_R[M]\} - \{K[M], V_R[N]\}. \quad (251)$$

Using the canonical convention

$$\{F, G\} = \int_{\Sigma} d^3x \left( \frac{\delta F}{\delta h_{ij}} \frac{\delta G}{\delta \pi^{ij}} - \frac{\delta F}{\delta \pi^{ij}} \frac{\delta G}{\delta h_{ij}} \right),$$

Eqs. (248) and (250) give

$$\begin{aligned} \{\mathcal{H}[N], \mathcal{H}[M]\} = 2AB \int_{\Sigma} d^3x & \left[ NP_{ij} (D^i D^j M - h^{ij} D^2 M) \right. \\ & \left. - MP_{ij} (D^i D^j N - h^{ij} D^2 N) \right]. \end{aligned} \quad (252)$$

In three spatial dimensions,

$$h^{ij} P_{ij} = (1 - 3\alpha)\pi. \quad (253)$$

Therefore

$$\begin{aligned} P_{ij} (D^i D^j M - h^{ij} D^2 M) &= \pi^{ij} D_i D_j M \\ &+ (2\alpha - 1)\pi D^2 M. \end{aligned} \quad (254)$$

Introduce

$$I_1[N, M] := \int_{\Sigma} d^3x (N\pi^{ij} D_i D_j M - M\pi^{ij} D_i D_j N), \quad (255)$$

$$I_2[N, M] := \int_{\Sigma} d^3x (N\pi D^2 M - M\pi D^2 N). \quad (256)$$

Then

$$\{\mathcal{H}[N], \mathcal{H}[M]\} = 2AB [I_1[N, M] + (2\alpha - 1)I_2[N, M]]. \quad (257)$$

Because  $\pi^{ij}$  is a weight-one density, covariant integration by parts may be used directly with the coordinate measure  $d^3x$ . For  $I_1$ ,

$$\begin{aligned} I_1[N, M] = - \int_{\Sigma} d^3x & \left[ (D_i N)\pi^{ij} D_j M + N(D_i \pi^{ij}) D_j M \right. \\ & \left. - (D_i M)\pi^{ij} D_j N - M(D_i \pi^{ij}) D_j N \right]. \end{aligned} \quad (258)$$

The first and third terms cancel because  $\pi^{ij} = \pi^{ji}$ . Hence

$$\begin{aligned} I_1[N, M] &= - \int_{\Sigma} d^3x (ND_j M - MD_j N) D_i \pi^{ij} \\ &= - \int_{\Sigma} d^3x \xi^i D_j \pi^j_i \\ &= \frac{1}{2} \mathcal{D}[\xi]. \end{aligned} \quad (259)$$

In the second line we used

$$ND_j M - MD_j N = h_{ij} \xi^i$$

and metric compatibility.

Similarly,

$$\begin{aligned} I_2[N, M] &= - \int_{\Sigma} d^3x \left( ND^i M - MD^i N \right) D_i \pi \\ &= - \int_{\Sigma} d^3x \xi^i D_i \pi. \end{aligned} \quad (260)$$

Combining these results yields

$$\begin{aligned} \{\mathcal{H}[N], \mathcal{H}[M]\} &= AB \mathcal{D}[\xi] \\ &\quad - 2AB(2\alpha - 1) \int_{\Sigma} d^3x \xi^i D_i \pi. \end{aligned} \quad (261)$$

The first term has the required hypersurface-deformation form. The second is an independent trace-gradient obstruction on the unrestricted metric phase space. For generic  $(h_{ij}, \pi^{ij})$  it is not proportional to the momentum constraint.

## B.2 Off-shell closure and ADM normalization

The target relation Eq. (241) contains a nonzero diffeomorphism generator for generic canonical data and smearing functions. A realization of that relation must therefore satisfy

$$AB \neq 0. \quad (262)$$

One could eliminate the obstruction in Eq. (261) on a restricted subspace by imposing, for example,

$$D_i \pi = 0. \quad (263)$$

No such additional phase-space condition belongs to the theory defined in Section 5. Requiring the target algebra to hold off shell for arbitrary canonical data, together with Eq. (262), therefore gives

$$2\alpha - 1 = 0, \quad (264)$$

and hence

$$\alpha = \frac{1}{2}. \quad (265)$$

The bracket then reduces to

$$\{\mathcal{H}[N], \mathcal{H}[M]\} = AB \mathcal{D}[\xi]. \quad (266)$$

Matching the normalization of Eq. (241) gives

$$AB = 1. \quad (267)$$

The coefficient  $C$  is not constrained by this calculation. Its contribution contains no derivatives of the smearing functions and therefore cancels in the antisymmetrized normal-normal bracket.

For completeness, the trace obstruction has a direct analogue in  $d > 1$  spatial dimensions. With the same quadratic kinetic ansatz,

$$h^{ij} P_{ij} = (1 - d\alpha)\pi, \quad (268)$$

and therefore

$$P_{ij} \left( D^i D^j M - h^{ij} D^2 M \right) = \pi^{ij} D_i D_j M + [(d-1)\alpha - 1] \pi D^2 M. \quad (269)$$

Provided the analogue of the target diffeomorphism term is nonzero, the trace obstruction vanishes off shell when

$$\alpha_d = \frac{1}{d-1}, \quad d > 1. \quad (270)$$

For  $d = 3$ , this gives Eq. (265). The case  $d = 1$  is degenerate and is not covered by this formula.

Returning to three spatial dimensions, the condition  $AB = 1$  fixes the relative normalization of the kinetic and curvature sectors. Introduce a nonzero constant  $\kappa$  by

$$A = 2\kappa, \quad B = \frac{1}{2\kappa}, \quad (271)$$

and write

$$C = \frac{\Lambda}{\kappa}. \quad (272)$$

Then

$$\mathcal{H}_\perp = \frac{2\kappa}{\sqrt{h}} \left( \pi^{ij} \pi_{ij} - \frac{1}{2} \pi^2 \right) - \frac{\sqrt{h}}{2\kappa} \left( {}^{(3)}R - 2\Lambda \right), \quad (273)$$

with

$$\mathcal{H}_i = -2D_j \pi^j_i. \quad (274)$$

These are the standard ADM constraints in the conventions of the main text.

The rescaling

$$A \mapsto \lambda A, \quad B \mapsto \lambda^{-1} B, \quad \lambda \neq 0, \quad (275)$$

preserves  $AB = 1$ . Thus hypersurface-deformation closure fixes only the relative kinetic–curvature normalization. It does not determine the overall gravitational scale. The conventional branch has

$$\kappa > 0.$$

As in Section 5, all canonical formulas in this appendix use  $c = 1$ . When conventional physical units are restored, the coupling appearing in the Einstein field equation is

$$\kappa_{\text{phys}} = \frac{8\pi G}{c^4}. \quad (276)$$

This empirical normalization is not determined by the closure calculation, and neither is the cosmological constant  $\Lambda$ .

The result of the appendix is the restricted implication

$$\begin{array}{ccc} \text{separable quadratic metric constraint} & & \\ + & & \\ \text{off-shell Lorentzian hypersurface-deformation} & \longrightarrow & \alpha = \frac{1}{2}, \\ \text{closure} & & AB = 1. \end{array} \quad (277)$$

The conclusion holds only within the ansatz stated in Section 5. It does not exclude other generally covariant canonical theories once curvature–momentum couplings, higher derivatives, additional fields, enlarged phase spaces, or different canonical representations are admitted.

## C Representative Covariant Checks

This appendix records two covariant checks of the local MS interpretation developed in Sections 2 and 3. No gravitational field equation is required in either calculation. The purpose is to verify that the bounded variables reproduce standard observer-relative massive-particle behavior while preserving

$$D^2 + S^2 = 1. \quad (278)$$

For a freely falling massive particle and an arbitrary observer congruence  $u^\mu$ , Section 2 gives

$$\frac{dS}{d\tau} = \frac{S^2}{c^2} U^\mu U^\nu \nabla_\mu u_\nu, \quad (279)$$

or, after decomposing the observer kinematics,

$$\frac{dS}{d\tau} = \frac{1}{c^2} \left( a_\mu v^\mu + \frac{\vartheta}{3} v^2 + \sigma_{\mu\nu} v^\mu v^\nu \right). \quad (280)$$

The vorticity does not contribute because  $v^\mu v^\nu \omega_{\mu\nu} = 0$ . The two examples below isolate, respectively, the expansion and acceleration terms in this equation.

### C.1 FLRW redshift in bounded MS variables

Consider a Friedmann–Lemaître–Robertson–Walker spacetime in cosmic time,

$$ds^2 = -c^2 dt^2 + a^2(t) \gamma_{ij} dx^i dx^j. \quad (281)$$

Let  $u^\mu$  be the comoving observer congruence. It is geodesic, shear free, and vorticity free,

$$a^\mu = 0, \quad \sigma_{\mu\nu} = 0, \quad \omega_{\mu\nu} = 0, \quad (282)$$

with expansion

$$\vartheta = 3H, \quad H := \frac{\dot{a}}{a}. \quad (283)$$

For a freely falling massive particle, Eq. (280) becomes

$$\frac{dS}{d\tau} = H \frac{v^2}{c^2}. \quad (284)$$

Let

$$D := \frac{|v|}{c}, \quad D^2 = 1 - S^2. \quad (285)$$

Then

$$\frac{dS}{d\tau} = H(1 - S^2). \quad (286)$$

For the comoving slicing,

$$N = 1,$$

and Eq. (43) gives

$$\frac{d\tau}{dt} = S. \quad (287)$$

Therefore

$$\frac{dS}{dt} = HS(1 - S^2). \quad (288)$$

For a noncomoving geodesic,  $D > 0$ , differentiation of Eq. (278) gives

$$\frac{dD}{dt} = -HD(1 - D^2). \quad (289)$$

Using

$$1 - D^2 = S^2, \quad 1 - S^2 = D^2,$$

Eqs. (288) and (289) imply

$$\begin{aligned} \frac{d}{dt} \ln\left(\frac{D}{S}\right) &= \frac{1}{D} \frac{dD}{dt} - \frac{1}{S} \frac{dS}{dt} \\ &= -HS^2 - HD^2 = -H. \end{aligned} \quad (290)$$

Hence

$$\frac{D}{S} = \frac{p}{a(t)}, \quad p > 0, \quad (291)$$

for every noncomoving geodesic. Here  $p$  is a constant of integration associated with the conserved comoving momentum scale.

The exactly comoving solution must be treated separately before taking the logarithm. It is

$$D(t) \equiv 0, \quad S(t) \equiv 1, \quad (292)$$

and is recovered continuously from the formulas below by setting  $p = 0$ .

Combining Eq. (291) with  $D^2 + S^2 = 1$  gives

$$S(t) = \frac{1}{\sqrt{1 + p^2/a^2(t)}}, \quad (293)$$

$$D(t) = \frac{p/a(t)}{\sqrt{1 + p^2/a^2(t)}}, \quad (294)$$

where now  $p \geq 0$ , with  $p = 0$  representing the comoving geodesic.

For a particle of rest mass  $m$ ,

$$Q = mc,$$

and the physical momentum magnitude measured by the comoving observer is

$$p = Q \frac{D}{S} = mc \frac{D}{S}. \quad (295)$$

Equation (291) therefore yields

$$p = \frac{mc}{a(t)} \propto a^{-1}. \quad (296)$$

This is the standard redshift of the physical momentum of a freely propagating massive particle relative to comoving observers.

In an expanding universe, every noncomoving freely propagating massive state therefore approaches

$$D \longrightarrow 0, \quad S \longrightarrow 1 \quad (297)$$

relative to the comoving congruence, while its momentum is encoded in the unbounded projective ratio  $D/S$ . The pointwise state surface is not deformed by the expansion; the geometry determines the observer-relative trajectory through that fixed state fiber.

## C.2 Schwarzschild static observers and the horizon

Consider the Schwarzschild exterior,

$$ds^2 = -N^2(r)c^2dt^2 + \frac{dr^2}{N^2(r)} + r^2d\Omega^2, \quad (298)$$

where

$$N(r) = \sqrt{1 - \frac{2GM}{c^2r}}. \quad (299)$$

For

$$r > \frac{2GM}{c^2},$$

let  $u^\mu$  be the static observer congruence,

$$u^\mu = \left( \frac{1}{N}, 0, 0, 0 \right). \quad (300)$$

With the normalization  $u^\mu u_\mu = -c^2$  used throughout this paper,  $u^\mu$  is the static observer four-velocity. The congruence has vanishing expansion and shear but nonzero proper acceleration. In its outward-pointing orthonormal radial direction,

$$a^{\hat{r}} = c^2 N'(r), \quad N'(r) := \frac{dN}{dr}. \quad (301)$$

For a radial timelike geodesic, define the conserved dimensionless Killing energy

$$\varepsilon := N^2 \frac{dt}{d\tau}. \quad (302)$$

The Lorentz factor relative to the static observer is

$$\gamma(U, u) = N \frac{dt}{d\tau} = \frac{\varepsilon}{N}, \quad (303)$$

and therefore

$$S = \frac{N}{\varepsilon}. \quad (304)$$

Let  $D$  denote the signed radial MS velocity, with  $D < 0$  on an infalling branch and  $D > 0$  on an outgoing branch. The radial velocity measured by the static observer is

$$v^{\hat{r}} = cD,$$

and

$$\frac{dr}{d\tau} = cN \frac{D}{S} = c\varepsilon D. \quad (305)$$

The second equality follows from Eq. (304).

Since

$$\vartheta = 0, \quad \sigma_{\mu\nu} = 0,$$

Eq. (280) gives

$$\frac{dS}{d\tau} = \frac{a_{\hat{r}} v^{\hat{r}}}{c^2} = cN'(r)D. \quad (306)$$

The same result follows by differentiating Eq. (304), using conservation of  $\varepsilon$  and Eq. (305).

To obtain the evolution of  $D$  without dividing by  $D$ , use the radial geodesic equation directly. The relevant Christoffel symbols are

$$\Gamma^r_{tt} = c^2 N^3 N', \quad \Gamma^r_{rr} = -\frac{N'}{N}. \quad (307)$$

Hence

$$\frac{d^2 r}{d\tau^2} + c^2 N^3 N' \left( \frac{dt}{d\tau} \right)^2 - \frac{N'}{N} \left( \frac{dr}{d\tau} \right)^2 = 0. \quad (308)$$

Using

$$\frac{dt}{d\tau} = \frac{\varepsilon}{N^2}$$

and the normalization of the timelike four-velocity,

$$\left( \frac{dr}{d\tau} \right)^2 = c^2 (\varepsilon^2 - N^2), \quad (309)$$

Eq. (308) reduces to

$$\frac{d^2 r}{d\tau^2} = -c^2 N N'. \quad (310)$$

Since  $dr/d\tau = c\varepsilon D$  and  $\varepsilon$  is constant,

$$\frac{dD}{d\tau} = -c \frac{N}{\varepsilon} N' = -c N'(r) S. \quad (311)$$

This derivation remains valid at a radial turning point  $D = 0$ . Away from such a point, the same relation also follows by differentiating  $D^2 + S^2 = 1$  and using Eq. (306).

Equations (306) and (311) preserve

$$D^2 + S^2 = 1$$

and describe the evolution of the signed pair  $(D, S)$  along the massive-state semicircle. Their position-dependent rate is controlled by the proper acceleration of the static observer congruence, rather than by a deformation of the local state geometry.

For a particle released from rest at infinity,

$$\varepsilon = 1. \quad (312)$$

Equation (304) then gives

$$S = \sqrt{1 - \frac{2GM}{c^2 r}}, \quad (313)$$

while the infalling branch of  $D^2 + S^2 = 1$  is

$$D = -\sqrt{\frac{2GM}{c^2 r}}. \quad (314)$$

Relative to the static congruence,

$$r \longrightarrow \frac{2GM}{c^2} \implies S \longrightarrow 0^+, \quad D \longrightarrow -1. \quad (315)$$

This does not mean that the massive geodesic becomes null. Its four-velocity continues to obey

$$g_{\mu\nu}U^\mu U^\nu = -c^2. \quad (316)$$

Rather, the static observer congruence fails to extend to the horizon as a regular timelike observer family. Correspondingly,

$$\gamma(U, u) = \frac{1}{S} \longrightarrow \infty, \quad (317)$$

so the static and freely falling frames become infinitely boosted relative to one another.

The value  $S = 0$  is not part of the physical massive state manifold. Its appearance here is entirely observer dependent. A regular freely falling observer crossing the same horizon may, for example, be instantaneously comoving with the particle and assign

$$D = 0, \quad S = 1$$

at the crossing event. The projective boundary of the bounded representation therefore carries no observer-independent implication that a massive trajectory has become null. As emphasized in Section 2, the separate null completion of the flat-space theory is a different construction.

The FLRW and Schwarzschild examples isolate two distinct terms in Eq. (280). FLRW evolution is driven by the expansion of the comoving congruence, whereas Schwarzschild static-observer evolution is driven by observer acceleration. In both cases

$$D^2 + S^2 = 1$$

remains a local kinematical identity. Spacetime geometry and the chosen observer congruence determine how a massive state moves through this bounded representation; they do not alter the pointwise massive-state surface itself.

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